Rational Planning and Self Consistency in Decision Problems with Anticipatory Emotions

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Abstract

Belief-dependent preferences capture psychological motivations such as emotions and image concerns. To manage certain emotions, some agents may prefer *not* to premeditate their behavior, but to remain undecided and act spontaneously. This raises conceptual and analytical challenges. We show that a rational agent never needs to act spontaneously if, at every history, his decision utility does not depend on what he planned to do at past or counterfactual histories. Decision problems with anticipatory emotions are a common class of psychological games where this property holds. To illustrate, we study the effects of savoring and dread on intertemporal choice. We find that behavior with and without commitment only coincides if the agent's emotions are very mild or very intense. Dynamic inconsistency arises for an intermediate range of emotional intensity.

Keywords: Psychological game theory, Rational planning, Spontaneous choice, Dynamic consistency, Anticipatory emotions, Savoring, Dread
JEL: C72, C73, D81, D91

1. Introduction

Belief-dependent preferences capture psychological motivations such as emotions and image concerns. These motivations influence human interactions and economic behavior. For instance, anger shapes voting preferences (Vasilopoulos et al., 2019), reciprocity affects bargaining strategies (Hoffman et al., 1998), and image concerns drive charitable behavior (Lacetera and Macis, 2010). Such incentives are typically modeled by allowing the agents' utility to depend on beliefs. For instance, picture a child receiving a book for his birthday. His satisfaction may well depend on what he expected to receive: he might be relieved if he expected a pair of socks, or disappointed if he expected a LEGO Dino Air Tracker.[1] Clearly, human well-being depends not only on the physical experience of material outcomes, but also on the conceptual consumption[2] of beliefs, information and mental imagery.

Anticipatory emotions are a common type of belief-dependent motivations, elicited in anticipation of future events. In this paper we are interested in savoring and dread, two particular types of anticipatory emotions. Savoring (resp. dread) is defined as the pleasant (resp. unpleasant) anticipation of future positive (resp. negative) events, and it generates an incentive to delay (resp. hasten) the experience

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[1] Any resemblance to actual events is purely coincidental.

[2] We borrow this terminology from Ariely and Norton (2009).

of positive (resp. negative) outcomes. These incentives oppose the standard predictions of discounted utility theory, according to which rational agents should prefer to experience immediately (resp. maximally delay) all positive (resp. negative) outcomes. For this reason, agents affected by savoring and dread display what is often referred to as "negative time preference" (Loewenstein and Prelec, 1991).

Loewenstein (1987) was the first to propose a formal model of savoring and dread suitable to economic analysis. Although Loewenstein's definitions are not explicitly belief-dependent, we reinterpret savoring and dread as belief-dependent motivations in a natural way: by defining them as the expectation of future material utility, we highlight their dependence on the agent's beliefs about the future. Loewenstein used his model to study a simple problem of intertemporal choice under commitment: a rational agent must choose when to experience a known outcome (up to an established deadline), and his initial choice is enforced by a commitment device. We wish to extend Loewenstein's analysis to intertemporal decision problems under discretion (i.e. without commitment), and compare the results. However, doing so is not exactly straightforward.

By allowing the agent to make choices at different points in time, we cause his anticipatory emotions to depend on his beliefs about his own behavior; we refer to these beliefs as the agent's plan. Indeed, the agent's emotions depend on his expectation of future outcomes. If the realization of future outcomes depends on his own future actions, his expectation of future outcomes must depend on his plan. This is not an innocuous observation. Since a decision problem is a degenerate game with perfect information, one might expect that a deterministic solution can always be found via "intrapersonal backward induction" (i.e. dynamic programming). However, this need not be the case: the literature on psychological game theory (see in particular Battigalli and Dufwenberg, 2009) documents that if the agent's utility depends on his own plan, a number of complications might arise.

Sometimes a rational plan fails to exist entirely, making the decision problem unsolvable. Sometimes there exists a unique rational plan, but it involves randomization and cannot be derived via backward induction;[3] this occurs when rational agents prefer *not* to premeditate their behavior, but to remain undecided and act spontaneously (see Section 2). Sometimes backward induction finds a deterministic rational plan, but fails to find other rational plans that yield strictly higher expected utility; and even if it finds a rational plan that induces no ties, it need not be unique. If all these examples seem counterintuitive, it is because they can never occur in standard game theory. One of our goals is to review exactly how own-plan dependence can lead to each of these oddities.

Despite everything we mentioned, if we actually try to study any decision problem with anticipatory emotions, none of these complications seem to arise. Upon closer inspection, we realize that this is due to the specific type of own-plan dependence exhibited by anticipatory emotions. In particular, the agent's anticipatory utility only depends on what he plans to do in the future, as opposed to what he planned to do at past or counterfactual histories. In the first half of the paper we formalize this property, which we call *past and counterfactual plan independence* (PCPI). We expand our analysis to general decision problems, and show that under PCPI a deterministic rational plan always exists, so rational agents

[3]All rational plans (if they exist) can be found with a fixed-point approach, but in practice this may be much harder than using a discrete algorithm such as backward induction.

never need to act spontaneously. Moreover, all rational plans can be found via backward induction.

In the second half of the paper we return to intertemporal decision problems with savoring and dread, where we apply our results to greatly simplify the analysis. We compare rational plans under commitment and discretion, and show that they only coincide if the agent's emotions are very mild or very intense. That is, dynamic inconsistency arises for an intermediate range of emotional intensity. We discuss in detail how dynamic inconsistency affects the strategic reasoning of a sophisticated agent, and why a full behavioral characterization of dynamic inconsistency is only possible in the case of dread.

Summing up, this paper consists of a purely theoretical part, where we present our main theorem, and a more applied part, where we apply the theorem to a specific decision problem. In the first half, we examine general properties of psychological motivations, and how they relate to notions of rationality, spontaneity, and self consistency. We present our theorem and compare it with related findings in the literature. In the second half, we apply our theorem to study the effects of savoring and dread on intertemporal choice, and provide a (partial) behavioral characterization of dynamic inconsistency.

The rest of the paper is organized as follows. Section 2 provides a more detailed introduction to plans, psychological preferences, and their properties. We review the literature and present a heuristic example. Section 3 summarizes the framework of psychological game theory in simplified form; experienced readers may wish to skim this part. Section 4 contains our main theorem, related results, and a range of examples illustrating the pitfalls of own-plan dependence. Section 5 introduces anticipatory emotions, savoring, and dread. We review the theoretical and experimental literature, and present a belief-dependent reinterpretation of Loewenstein's model. Readers primarily interested in applied theory or anticipatory emotions may skip directly to Section 5 without losing essential details. Finally, Section 6 contains our applied theory results.

2. Plans, psychological preferences, and heuristic examples.

Psychological game theory (PGT) is a sub-field of game theory where players' utility is allowed to depend on beliefs. PGT was pioneered by Geanakoplos et al. (1989) and further developed by Battigalli and Dufwenberg (2009), with important refinements in Battigalli et al. (2019a). For a comprehensive survey of recent developments, see Battigalli and Dufwenberg (2022). In modern PGT, a player's psychological utility may depend on various types of beliefs:

- Own beliefs about:
  1. Nature (i.e. players' traits and chance moves)
  2. Co-players' beliefs and behavior
  3. Own behavior
- Co-players' beliefs about anyone's traits, beliefs and behavior.

Beliefs about nature are exogenous: players are simply endowed with them. Beliefs about co-players' beliefs and behavior are endogenous: they derive from interactive reasoning, such as forward or backward induction. Finally, beliefs about own behavior (i.e. plans[4]) are intentional: players believe they will act a

[4]Whether this is an accurate definition of plans is debated in the philosophical literature. Velleman (1989) defends this view, while Bratman (1987) argues against it; see Velleman (1991) for a critical review.

certain way because they *intend* to act that way. The special nature of plans explains why complications might arise when a player's utility depends on his own plan. To see precisely what can go wrong, we first need a clearer understanding of plans as mathematical objects.

2.1. Plan (in)dependence.

Each player holds a single plan, that prescribes how to behave at every node of the game. Formally, a plan maps every non-terminal history to a probability measure over feasible actions.[5] Plans are a choice variable: each player formulates his own plan, and the act of planning is carried out entirely before the game begins.[6] Beliefs about co-players and nature logically precede beliefs about own behavior. A well-reasoning agent first consults his beliefs about others, and only then constructs his own plan in response to those beliefs.

The actual behavior of every agent is modeled as an external (non-mental) state of the world, which *may or may not* coincide with the agent's plan.[7] A plan is rational if holding it does not elicit an incentive to deviate from it.[8] A plan is deterministic (or pure) if it ascribes full probability to a single action per history, and probabilistic (or mixed) otherwise.

Enacting a deterministic plan is fully intentional: one acts deliberately, having planned his contingent behavior down to every detail. Enacting a probabilistic plan is seen as "less" intentional: according to the standard interpretation, one correlates his behavior with the (utility-irrelevant) realization of an extraneous random variable. This interpretation is reasonable but abstract. In the real world, people often hold probabilistic beliefs about their own future behavior; however, we argue that this rarely reflects the intention to flip a coin or roll a die, and more often conveys inability (or unwillingness) to decide in advance.

Therefore, we adopt a different interpretation of mixed plans.[9] An agent holding a mixed plan intends to select an action from the support of his plan, but remains undecided among all actions in the support. The plan represents his probabilistic assessment of his own future behavior. When an action is finally played, we view as intentional the fact that it belongs to the plan's support; however, the selection between different elements of the support is seen as a "spontaneous," unintentional act.[10] In other words, the agent serves as his own randomization device.

This weak form of intentionality is *not* at odds with rationality. Indeed, in some cases, remaining

[5]In the language of standard game theory, plans would be called behavior strategies. However, conflating the two concepts is misleading: plans are beliefs, not behavior.

[6]For an alternative framework where plans can be revised mid-game, see Battigalli et al. (2023).

[7]In this paper we focus only on the planning stage, so we remain agnostic about the consistency between the agent's plan and behavior. See Battigalli and De Vito (2021) for a detailed discussion of this issue.

[8]When specialized to the framework of Kőszegi and Rabin (2006, 2009) and Kőszegi (2010), our notion of rational plans coincides with their notion of personal equilibrium (PE). Therefore, all of our claims about rational plans can be interpreted in the language of Kőszegi and Rabin as claims about PE.

[9]This interpretation is conceptually different, but mathematically identical; choosing one is a matter of taste.

[10]We purposefully avoid the term "impulsive." Both "impulsive" and "spontaneous" convey the idea of unplanned and unintentional behavior, but "impulsive" suggests that the action is triggered by an emotional impulse. Such impulses are usually modeled as utility distortions, which we ignore in this paper.

undecided may be the only rational option. This is a particularly interesting consequence of own-plan dependence.

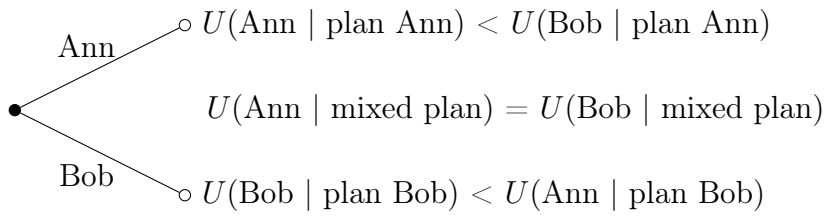


Figure 1: Heuristic example.

Heuristic example (guilt from premeditation).
Consider the example of Figure 1. The agent has a single ticket for a concert he cannot attend, and wants to give it to one of his friends (Ann or Bob). He has no intrinsic preference between his friends, so he enjoys the same altruistic utility by giving the ticket to either one. However, he experiences guilt toward whomever he leaves without ticket. His guilt is amplified by premeditation: the more he plans to leave someone without ticket, the guiltier he feels by doing so. If he plans to give the ticket to Ann (resp. Bob), doing so would make him feel much guiltier than giving it to Bob (resp. Ann): therefore, he actually prefers to give the ticket to Bob (resp. Ann). His utility depends on his plan in such a way that $U(\text{Ann} \mid \text{plan Ann}) < U(\text{Bob} \mid \text{plan Ann})$ and $U(\text{Bob} \mid \text{plan Bob}) < U(\text{Ann} \mid \text{plan Bob})$.

Consequently, there is no deterministic rational plan. Whichever deterministic plan the agent holds, he has a strict incentive to deviate from it. However (under a mild continuity assumption), a probabilistic rational plan does exist, and it satisfies $U(\text{Ann} \mid \text{mixed plan}) = U(\text{Bob} \mid \text{mixed plan})$. Holding the rational plan makes the agent indifferent between the two actions, so his utility is maximized regardless of what he ends up doing spontaneously. In other words, the agent's dilemma torments him so much that he cannot make up his mind. The only way to manage his guilt is to remain undecided between disappointing Ann or Bob. This way, he feels "less responsible" for the final outcome, which mitigates his guilt.[11] □

The rational agent is forced to act spontaneously, rather than fully premeditate his behavior. This is not a pathology of the framework: many natural examples follow the same pattern. However, this complicates the analysis for three main reasons. The first reason is conceptual: often, we capture the way in which (one or more) players reason through a game of perfect information by solving the game with a folding-back algorithm. Under certain forms of own-plan dependence, this algorithm may fail, because its early steps depend on the output of later ones, leading to circular reasoning. The second reason is computational: if we cannot focus on deterministic plans WLOG, the set of candidate rational plans becomes uncountable. Proceeding by elimination is impossible, and the only way to find all rational plans is to solve a (possibly complicated) fixed-point problem.

[11] This example resembles the Machina's Mother (1989) story. The difference is that Machina's Mother *strictly* prefers to play whichever action is prescribed by her coin. This is only possible if the coin itself is a utility-relevant object: Machina's Mother derives added utility (e.g. a sense of fairness) by doing exactly what the coin says.

The final reason concerns behavioral predictions. In general, it is very hard to know exactly what is going to happen in a game. The accuracy of our predictions depends on the complexity of the game, and on the solution concept of our choice. However, there are some simple settings where we should be able to know what to expect. For instance, in standard game theory, games of perfect information without ties admit a unique solution, typically found by backward induction. However, in psychological game theory, this may not be the case. The non-existence of deterministic rational plans makes behavioral predictions harder to formulate and test: how can we predict what the agent will do, if he himself cannot?

Much of the PGT literature circumvents these complications by assuming own-plan independence (OPI): no player's utility can depend on his own plan. Under this assumption, it is well-known that a deterministic rational plan exists for every player. Moreover, all rational plans yield the same expected utility, and every player is dynamically consistent. This is a very useful result, but it comes at a cost. First, OPI is a stringent requirement, and many common emotions such as savoring and dread do not satisfy it. Second, OPI gets us more than what we need: it guarantees both deterministic rational planning *and* dynamic consistency. Since these properties are not logically related, in principle we should be able to secure either one without necessarily securing the other.

Our ultimate goal is not to eliminate the complications induced by own-plan dependence, but to provide a fundamental criterion to assess when they do not occur. Ideally, we seek a property that ensures the existence of deterministic rational plans, but is weaker than OPI in that it allows for own-plan dependence and dynamic inconsistency. We propose one such property, which we call "past and counterfactual plan independence" (PCPI): at every history, each player's decision utility[12] cannot depend on what he planned to do at past or counterfactual histories; however, it may depend on his plans for the future.

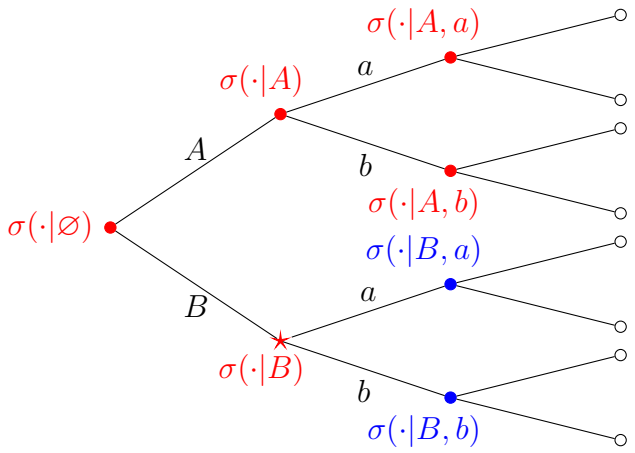


Figure 2: Illustration of PCPI in a single-player decision problem.

See Figure 2 for an illustration of the property. At history $\star$, the continuation value of playing a or b is only allowed to depend on future components of the plan, not past or counterfactual. The same restriction must hold at *every* history of the game. PCPI ensures that all rational plans (and in particular

[12]With "decision utility" we mean the continuation value of playing any given action. See details in Section 3.

a deterministic one) can be constructed in the intuitive folding-back fashion, starting from the longest histories of the game and planning backward. Early steps of the algorithm can no longer depend on the output of later steps, because later steps involve histories that are past or counterfactual to those involved in early steps.

Throughout the paper, we restrict our attention to decision problems, i.e. degenerate games with a single personal player (the agent) and chance. This is only to simplify exposition: all of our results can be generalized to multi-player games with incomplete information and perfect recall. Still, there is an important feature of our framework that cannot be simplified away: we cannot get rid of chance, because we need at least one co-player (albeit impersonal) to justify the "counterfactual" part of PCPI. Indeed, in decision problems without chance, PCPI collapses to simple "past plan independence" (see Section 4.4).

Lastly, note that PCPI is sufficient but not necessary for the existence of deterministic rational plans. As a sufficient property, its usefulness depends on how easily it can be checked and how prevalent it is in applications. We try to make PCPI as intuitive and straightforward as possible; indeed, we could consider weaker sufficient properties, "closer to necessity," but they would be far more cumbersome to state and analyze. Actually, we could achieve the same results with a stronger property (PBI, see Section 4.4), but we focus on PCPI as it is a natural counterpart to the literature's standard (OPI). Finally, decision problems with anticipatory emotions are a prominent class of psychological games where PCPI holds, but OPI does not. Anticipatory emotions arise frequently in applications, and play a central role in problems of intertemporal choice.

3. Theoretical framework.

Let X be any set, and (X_n) any sequence of sets. Let J, K be arbitrary integers with $0 < J \leq K$. Let x be a real number, and (x_n) a sequence of real numbers. Let f be an arbitrary random variable on $(X, 2^X)$, and $p \in \Delta(X)$ be an arbitrary probability measure on 2^X. Table 1 summarizes our notation.

Symbol	Meaning	Symbol	Meaning	Symbol	Meaning
0^0	1	$\mathbf{1}_X$	Indicator function	$[x]^+$	$\max\{x, 0\}$
$\varnothing$	Empty sequence	X^0	$\{\varnothing\}$	$[x]^-$	$\min\{x, 0\}$
$\emptyset$	Empty set	$X_\varnothing$	$X \cup \{\varnothing\}$	$\sum_{n\in\emptyset} x_n$	0
$\mathbb{N}$	$\{1, 2, \cdots\}$	$X^{\leq K}$	$\bigcup_{k=1}^{K} X^k$	$\prod_{n\in\emptyset} x_n$	1
$\mathbb{N}_0$	$\mathbb{N} \cup \{0\}$	$X^{J\leq K}$	$\bigcup_{k=J}^{K} X^k$	$(x_n)_{n\in\emptyset}$	$\varnothing$
$\mathbb{R}_+$	$\{x' \in \mathbb{R} \colon x' \geq 0\}$	$[x, \infty]$	$[x, \infty) \cup \{\infty\}$	$\times_{n\in\emptyset} X_n$	$\{\varnothing\}$
$\mathbb{R}_{++}$	$\{x' \in \mathbb{R} \colon x' > 0\}$	$\mathbb{E}[f; p]$	$\int f \mathrm{d}p$	$\bigcup_{n\in\emptyset} X_n$	$\emptyset$

Table 1: Mathematical notation.

3.1. Decision tree.

We work toward the definition of an intertemporal decision tree. We assume perfect and complete information.

Players. Let $\mathcal{I} = \{i, c\}$ be the set of players, with generic element ι. For all $\iota \in \mathcal{I}$, let $-\iota := \mathcal{I} \setminus \{\iota\}$. Player i is the only personal player, known as "the agent." Player c represents chance, or nature; we use the two terms interchangeably. We adopt a deterministic perspective: there is no objective probability, and all stochastic objects simply represent the agent's uncertainty.

Periods. Every decision problem is divided into a number of periods, $T \in \mathbb{N}$, which coincides with the horizon of the game. Let $\mathbb{T} := \{1, ..., T\} \subset \mathbb{N}$ be the set of all periods. Each period is an interval of time between consecutive dates, so $\mathbb{T}_0 := \mathbb{T} \cup \{0\}$ is the set of all dates. Each date $t \in \mathbb{T}$ occurs at the very end of period t; date 0 occurs at the start of period 1.

Outcomes. At the end of each period, the agent receives a material outcome. Let Y be the (finite) set of potentially feasible material outcomes, and $Y^{\leq T}$ be the set of potentially feasible streams of material outcomes, with generic element $y^k \equiv (y_\tau)_{\tau=1}^k$. Receiving nothing can be modeled as receiving the null outcome $\aleph$.

Actions. Each period is composed of a fixed number of stages, $L \in \mathbb{N}$. Both players act simultaneously at every stage. Let A_ι be the (finite) set of potentially feasible actions for player $\iota \in \mathcal{I}$, and $\mathbf{A} := A_i \times A_c$ be the set of potentially feasible action pairs, with generic element $\mathbf{a}$. Inactive players can be modeled as playing the waiting action w.

Histories. A history is a sequence of action pairs. The empty history $\varnothing$ is where the decision tree begins. Let $\widehat{H} \subseteq \mathbf{A}_{\varnothing}^{\leq LT}$ be the set of all feasible histories, which is closed under the weak "prefix of" relation $\preceq$. We partition $\widehat{H}$ into Z and H, respectively the set of terminal and non-terminal histories. History z is terminal iff there exists no $h \in \widehat{H}$ such that $z \prec h$.

Properties of histories. The set of histories belonging to period 1 is $\widehat{H} \cap \mathbf{A}_{\varnothing}^{1 \leq L}$. For all $t > 1$, the set of histories belonging to period t is $\widehat{H} \cap \mathbf{A}^{L(t-1)+1 \leq Lt}$. Let $\vec{H} := \widehat{H} \cap \bigcup_{t \in \mathbb{T}} \mathbf{A}^{Lt}$ be the set of all histories that occur at the end of some period. For notational convenience, we assume that a decision problem cannot terminate in the middle of a period, but only at the end.[13] In other words, we assume $Z \subseteq \vec{H}$. For every history $h \in \widehat{H}$, let $\lambda(h) \in \mathbb{N}_0$ be the length of h. Formally, $\lambda(h) = n$ iff $h \in \mathbf{A}^n$. Finally, let $\mathbf{T} \colon \widehat{H} \to \mathbb{T}$ be a function that maps each history to the period it belongs to:

$$\mathbf{T}(h) := \begin{cases} 1 & \text{if} \quad h = \varnothing \\ \left\lceil \dfrac{\lambda(h)}{L} \right\rceil & \text{if} \quad h \neq \varnothing \end{cases}$$

Feasibility. Let $\mathcal{A}_\iota \colon \widehat{H} \rightrightarrows A_\iota$ be the feasibility correspondence of player $\iota \in \mathcal{I}$, defined as $\mathcal{A}_\iota(h) := \{a_\iota \in A_\iota \colon \exists a_{-\iota} \in A_{-\iota},\ (h, (a_\iota, a_{-\iota})) \in \widehat{H}\}$. Let $\boldsymbol{\mathcal{A}}(h) := \mathcal{A}_i(h) \times \mathcal{A}_c(h)$. Notice that $z \in Z$ iff $\mathcal{A}_i(z) = \mathcal{A}_c(z) = \emptyset$. Player ι is inactive at history h if $\mathcal{A}_\iota(h) = \{w\}$. For all $\iota \in \mathcal{I}$, $h \in H$, and $z \in Z$ with $z \succ h$, let $\alpha_\iota(h, z)$ be the only action in $\mathcal{A}_\iota(h)$ that is compatible with z. Formally, $\alpha_\iota(h, z) = a_\iota$ iff $z \succ h$ and there exists some $a_{-\iota} \in A_{-\iota}$ such that $(h, (a_\iota, a_{-\iota})) \preceq z$.

[13] This is without loss of generality, as both players can be forced to take the waiting action at each remaining stage of a given period.

Outcome functions. Let $\widehat{\pi}\colon \vec{H} \to Y$ be the instantaneous outcome function, which yields the material outcome received by the agent at history $h \in \vec{H}$. Let $\pi\colon \vec{H} \to Y^{\leq T}$ be the cumulative outcome function, which yields the stream of material outcomes received by the agent up to history $h \in \vec{H}$. Formally, $\pi(h) := (\widehat{\pi}(h'))_{h' \preceq h}^{h' \in \vec{H}}$.

An intertemporal decision tree is a tuple $\left\langle T, \widehat{H}, (A_\iota)_{\iota \in \mathcal{I}}, Y, \pi \right\rangle$.

We illustrate the framework up to this point by introducing our running example.

Running example (decision tree).
Consider Figure 3. In the first period, the agent is invited to a restaurant. Initially, he does not know whether the restaurant serves meat or fish. At this point, the agent and chance are both inactive. In the second period, the agent goes to the restaurant and discovers the menu: we model this as a chance move. No material outcome is received in the first two periods. In the third period, the agent can choose between a cold appetizer (liver pâté or crab salad) or a warm entrée (grilled beef or roasted bass).

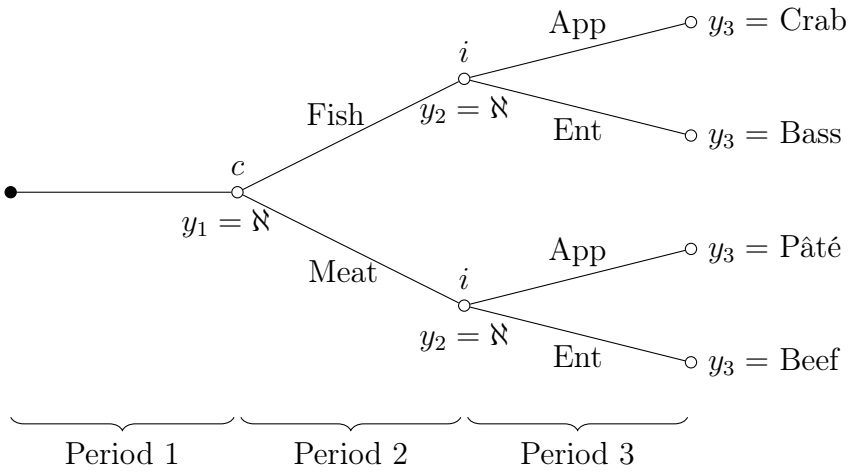


Figure 3: Running example.

There are three periods and four dates, so $T = 3$. The set of outcomes is $Y = \{\text{Beef, Pâté, Crab, Bass, } \aleph\}$. The sets of actions are $A_i = \{\text{Ent, App, } w\}$ and $A_c = \{\text{Meat, Fish, } w\}$. Stages and periods coincide, so $L = 1$ and $\vec{H} = \widehat{H} \setminus \{\varnothing\}$. To simplify notation, let $h_1 = (w, w)$, and:

$$h_2^{\mathbf{f}} := (w,w),(w,\text{Fish}) \qquad h_2^{\mathbf{m}} := (w,w),(w,\text{Meat})$$
$$h_3^{\mathbf{fa}} := (w,w),(w,\text{Fish}),(\text{App},w) \qquad h_3^{\mathbf{fe}} := (w,w),(w,\text{Fish}),(\text{Ent},w)$$
$$h_3^{\mathbf{ma}} := (w,w),(w,\text{Meat}),(\text{App},w) \qquad h_3^{\mathbf{me}} := (w,w),(w,\text{Meat}),(\text{Ent},w)$$

The numerical indices in our simplified history notation indicate both the length of the history and the period to which it belongs. The feasibility correspondences are $\mathcal{A}_i(\varnothing) = \mathcal{A}_c(\varnothing) = \mathcal{A}_i(h_1) = \mathcal{A}_c(h_2^{\mathbf{f}}) = \mathcal{A}_c(h_2^{\mathbf{m}}) = \{w\}$, $\mathcal{A}_i(h_2^{\mathbf{f}}) = \mathcal{A}_i(h_2^{\mathbf{m}}) = \{\text{App}, \text{Ent}\}$ and $\mathcal{A}_c(h_1) = \{\text{Meat}, \text{Fish}\}$. The cumulative outcome

function is:

$$\pi(h_3^{\mathbf{fa}}) = (\aleph, \aleph, \text{Crab}) \qquad \pi(h_3^{\mathbf{fe}}) = (\aleph, \aleph, \text{Bass})$$
$$\pi(h_3^{\mathbf{ma}}) = (\aleph, \aleph, \text{Pâté}) \qquad \pi(h_3^{\mathbf{me}}) = (\aleph, \aleph, \text{Beef})$$

3.2. Utility functions.

We now turn to the definition of a psychological utility function. We assume that the agent's preferences do not depend on features of the decision tree.[14] This maintains a formal and conceptual separation between the agent and the "rules of the game." Relaxing this assumption would be necessary to model game-dependent emotions such as regret.

Since our focus is on belief-dependent sources of dynamic inconsistency, we abstract away from other sources of inconsistency that might interfere with the analysis. Consequently, we rule out quasi-hyperbolic discounting and utility distortions.[15]

Personal traits. The agent's traits are an array of utility-relevant parameters representing known personal features such as skills, intelligence, or emotional sensitivity. Let Θ be the set of all potential traits, with generic element θ.

Material beliefs. At every stage of the decision problem, the agent holds probabilistic beliefs about the (entire) stream of material outcomes he is bound to receive. Let $M := \Delta(Y^{\leq T})$ be the space of such material beliefs. We restrict the agent's preferences to depend on beliefs held at every date of the decision problem, rather than at every stage.[16] Therefore, we let $M^{\leq T+1}$ be the space of utility-relevant streams of material beliefs, with generic element $\mu_0^k \equiv (\mu_\tau)_{\tau=0}^k$.

Instantaneous utility. For all $t \in \mathbb{T}$, let $\nu_t \colon Y^t \times \Theta \times M^{t+1} \to \mathbb{R}$ be the period-t instantaneous psychological utility function of the agent. It depends on:

1. The stream of outcomes received by the agent up to date t: $y^t \in Y^t$;
2. The agent's traits: $\theta \in \Theta$;
3. The stream of material beliefs held by the agent up to date t: $\mu_0^t \in M^{t+1}$.

This function yields the utility experienced by the agent at date t: not sooner, nor later.

Intertemporal utility. For all $t \in \mathbb{T}$, let $V_t \colon (Y \times M)^{t \leq T} \times M \times \Theta \to \mathbb{R}$ be the period-t intertemporal psychological utility function of the agent. It is defined as $V_t(y^k, \mu_0^k, \theta) := \sum_{\tau=t}^k \delta^{\tau-t} \nu_\tau(y^\tau, \theta, \mu_0^\tau)$, where $\delta \in [0, 1]$ is an intertemporal discount factor and $t \leq k \in \mathbb{T}$. We treat δ as one of the agent's traits. Function V_t represents the agent's intertemporal psychological preferences in period t. Since V_t depends on variables that only realize after period t, the agent consults his current beliefs to "compute" its expectation, which is what he ultimately aims to maximize if he is rational.

[14] This is another simplifying assumption that could be relaxed without affecting our results.

[15] As explained in Battigalli et al. (2019a), utility distortions represent visceral urges arising from "heated" emotions, such as the impulse to harm others when feeling angry (Battigalli et al., 2019b).

[16] This is without loss of generality: by setting $L = 1$, stages and periods coincide, and all beliefs become utility-relevant.

An intertemporal decision problem is a tuple $\left\langle T, \widehat{H}, (A_\iota)_{\iota\in\mathcal{I}}, \Theta, Y, \pi, (V_t)_{t\in\mathbb{T}} \right\rangle$.

Behavioral beliefs. The agent's plan specifies how he intends to behave at every history $h \in H$. The space of plans is $\Delta^i := \times_{h\in H}\Delta(\mathcal{A}_i(h))$, with generic element $\sigma^i \equiv (\sigma^i(\cdot|h))_{h\in H}$. The agent's conjecture specifies how he believes chance will behave at every history $h \in H$. The space of conjectures is $\Delta^c := \times_{h\in H}\Delta(\mathcal{A}_c(h))$, with generic element $\sigma^c \equiv (\sigma^c(\cdot|h))_{h\in H}$. For all $\iota \in \mathcal{I}$, $\sigma^\iota \in \Delta^\iota$, and $h \in H$, we refer to $\sigma^\iota(\cdot|h)$ as the h-component of σ^ι. Let $\boldsymbol{\Delta} := \Delta^i \times \Delta^c$ be the space of behavioral beliefs, with generic element $\boldsymbol{\sigma} = (\sigma^i, \sigma^c)$.

Path beliefs. From the agent's behavioral beliefs, we can derive his contingent beliefs about the terminal history he is bound to reach. Let $\Delta(Z)$ be the space of such beliefs. For all $\boldsymbol{\sigma} = (\sigma^i, \sigma^c) \in \boldsymbol{\Delta}$ and $h \in \widehat{H}$, let $\mathbb{P}^h_{\boldsymbol{\sigma}} \in \Delta(Z)$ be the path belief induced by $\boldsymbol{\sigma}$ at history h. Formally, $\mathbb{P}^h_{\boldsymbol{\sigma}}(z) := \prod_{h\preceq h'\prec z}[\sigma^i(\alpha_i(h',z)|h') \cdot \sigma^c(\alpha_c(h',z)|h')] \cdot \mathbf{1}_{\{z'\in Z\colon z'\succeq h\}}(z)$. Notice that $\mathbb{P}^z_{\boldsymbol{\sigma}}(z) = 1$ for all $z \in Z$ and $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$ (using Table 1, row 4 column 3).

Belief functions. Let $\widehat{\psi}\colon \vec{H}_\varnothing \times \boldsymbol{\Delta} \to M$ be the instantaneous belief function, which yields the material belief held by the agent at history $h \in \vec{H}_\varnothing$, given his behavioral beliefs $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$. Formally, $\widehat{\psi}(h, \boldsymbol{\sigma})(y^k) := \mathbb{P}^h_{\boldsymbol{\sigma}}(\{z \in Z\colon \pi(z) = y^k\})$. Let $\psi\colon \vec{H}_\varnothing \times \boldsymbol{\Delta} \to M^{\leq T+1}$ be the cumulative belief function, which yields the stream of utility-relevant material beliefs held by the agent up to history $h \in \vec{H}_\varnothing$, given his behavioral beliefs $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$. Formally, $\psi(h, \boldsymbol{\sigma}) := (\widehat{\psi}(h', \boldsymbol{\sigma}))^{h'\in\vec{H}_\varnothing}_{h'\preceq h}$.

Path utility. For all $t \in \mathbb{T}$, let $U_t\colon Z \times \Theta \times \boldsymbol{\Delta} \to \mathbb{R}$ be the period-t path utility function of the agent.[17] It maps each terminal history $z \in Z$, vector of traits $\theta \in \Theta$, and beliefs $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$ to the intertemporal psychological utility they induce at date t. Formally, $U_t(z, \theta, \boldsymbol{\sigma}) := V_t(\pi(z), \psi(z, \boldsymbol{\sigma}), \theta)$.

Decision utility. Let $\widehat{U}\colon \{(a_i, h) \in A_i \times H\colon a_i \in \mathcal{A}_i(h)\} \times \Theta \times \boldsymbol{\Delta} \to \mathbb{R}$ be the decision utility function of the agent.[18] It yields the expected value of playing action $a_i \in \mathcal{A}_i(h)$ at history $h \in H$, given traits $\theta \in \Theta$ and beliefs $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$. Formally:

$$\widehat{U}(a_i, h, \theta, \boldsymbol{\sigma}) := \sum_{a_c\in\mathcal{A}_c(h)} \sigma^c(a_c|h) \cdot \left[\sum_{z\in Z} U_{\mathbf{T}(h,(a_i,a_c))}(z, \theta, \boldsymbol{\sigma}) \cdot \mathbb{P}^{h,(a_i,a_c)}_{\boldsymbol{\sigma}}(z)\right]$$

Continuation utility. Let $\vec{U}\colon \widehat{H} \times \Theta \times \boldsymbol{\Delta} \to \mathbb{R}$ be the continuation utility function of the agent. It yields the expected value of starting the game at history $h \in \widehat{H}$, given traits $\theta \in \Theta$ and beliefs $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$. Formally:

$$\vec{U}(h, \theta, \boldsymbol{\sigma}) := \begin{cases} \sum\limits_{a_i\in\mathcal{A}_i(h)} \sigma^i(a_i|h) \cdot \widehat{U}(a_i, h, \theta, \boldsymbol{\sigma}) & \text{if} \quad h \in H \\ U_{\mathbf{T}(h)}(h, \theta, \boldsymbol{\sigma}) & \text{if} \quad h \in Z \end{cases}$$

Notice that the instantaneous and intertemporal utility functions can be defined without any knowledge of the decision tree: as anticipated, the agent's psychological preferences do not depend on the rules

[17] In the language of standard game theory, this is usually called the "payoff function." We avoid this term due to its potential ambiguity.

[18] Notice that decision utility at history $h \in \vec{H}$ does *not* include the instantaneous utility experienced at history h.

of the game. Conversely, the path, decision, and continuation utility functions are hybrid objects that combine features of the game (e.g. actions and histories) with features of the agent (e.g. psychological preferences).

We continue to illustrate the framework using our running example.

Running example (utility functions).
It is convenient to partition Y into three cells (entrées, appetizers, and nothing): $Y = \{\text{Bass, Beef}\} \cup \{\text{Crab, Pâté}\} \cup \{\aleph\}$. For all $y \in Y$, let $[y]$ be the cell containing y. Let $\mathbf{App} = [\text{Crab}]$ and $\mathbf{Ent} = [\text{Bass}]$. Let $\Theta = \{\theta\}$ for some $\theta \in \mathbb{R}^7$. Let $m\colon Y \times \mathbb{R} \to \mathbb{R}$ be the agent's material utility function, which captures his taste for different kinds of dishes. Naturally, $m(\aleph, \cdot) = 0$. Suppose that the agent likes meat and dislikes fish; his preference for different courses is parametrized by θ_1. Formally:

$$\begin{aligned} m(\text{Pâté}, \theta_1) &= 1 + \theta_1 & m(\text{Beef}, \theta_1) &= 1 \\ m(\text{Crab}, \theta_1) &= -1 + \theta_1 & m(\text{Bass}, \theta_1) &= -1 \end{aligned}$$

We now specify the instantaneous utility function as:

$$\begin{aligned} \nu_t(y^t, \theta, \mu_0^t) = {} & \underbrace{m(y_t, \theta_1)}_{\text{material utility}} + \theta_2 \cdot \underbrace{\mathbf{1}_{\{0\}}\left(\mu_{t-1}\left(Y^{t-1} \times \{y_t\} \times Y^{3-t}\right)\right)}_{\text{surprise}} \\ & + \theta_3 \cdot \underbrace{\left[\frac{1 - \mathbf{1}_{\{\aleph\}}(y_t)}{1 + \mathbf{1}_{\mathbf{App}}(y_t)}\right]}_{\text{hunger effect}} \cdot \underbrace{\mu_{t-1}\left(Y^{t-1} \times \arg\max_{y \in Y} m(y, \theta_1) \times Y^{3-t}\right)^2}_{\text{appetite}} \\ & + \theta_4 \cdot \underbrace{\left[-\mathbf{1}_{\{1,2\}}(t) \cdot \sum_{y^3 \in \text{ supp } \mu_t} \left(\mu_t(y^3) - \frac{1}{\max\{|\text{supp } \mu_t|, 2\}}\right)^2\right]}_{\text{belief uniformity}} \\ & + \theta_5 \cdot \underbrace{\mu_{t-1}\left(Y^{t-1} \times (Y \setminus \{y_t\}) \times Y^{3-t}\right)}_{\text{outcome unlikeliness}} \\ & + \theta_6 \cdot \underbrace{\left[-\mathbf{1}_{\{1,2\}}(t) \cdot \mu_t\left(Y^t \times \arg\max_{y \in Y} m(y, \theta_1) \times Y^{2-t}\right)\right]}_{\text{impatience}} \\ & + \theta_7 \cdot \underbrace{m(y_t, \theta_1) \cdot \mu_0\left(Y^{t-1} \times ([y_t] \setminus \{y_t\}) \times Y^{3-t}\right)}_{\text{reduced-form relief / distress}^{19}} \end{aligned}$$

To avoid confusion, keep in mind that material beliefs ascribe probabilities not to individual outcomes, but to entire streams of material outcomes (which in this case all belong to Y^3). We will clarify the interpretation of different terms of the utility function as we study how they affect rational planning and self consistency. For now, note that $\theta = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7)$ is a vector of parameters that represent the agent's taste for different courses, as well as his attitude (aversion vs propensity) and sensitivity to

[19] "Reduced-form" signifies that we are mimicking the effects of relief and distress without appealing to a general theory.

different psychological motivations.

For simplicity, fix $\delta = 1$ throughout the running example. To ease notation, we omit δ from the vector of traits. This explains why we let $\theta \in \mathbb{R}^7$. From now on, suppose that the agent's conjecture σ^c assigns equal probability to the restaurant serving meat or fish, so $\sigma^c(\text{Meat}|h_1) = \sigma^c(\text{Fish}|h_1) = 1/2$. Then, the continuation utility computed at h_1 becomes:

$$\begin{aligned}
\vec{U}(h_1, \theta, \boldsymbol{\sigma}) = \theta_1 \cdot & \underbrace{\frac{1}{2}[\sigma^i(\text{App}|h_2^{\mathbf{m}}) + \sigma^i(\text{App}|h_2^{\mathbf{f}})]}_{\text{expected material utility}} \\
+ \theta_3 \cdot & \underbrace{\left[\frac{1}{2}\sigma^i(\text{App}|h_2^{\mathbf{m}}) + \sigma^i(\text{Ent}|h_2^{\mathbf{m}})\right]}_{\text{average hunger effect}} \\
& \cdot \underbrace{\frac{1}{2}\left[\mathbf{1}_{[0,\infty)}(\theta_1) \cdot \sigma^i(\text{App}|h_2^{\mathbf{m}}) + \mathbf{1}_{(-\infty,0]}(\theta_1) \cdot \sigma^i(\text{Ent}|h_2^{\mathbf{m}})\right]^2}_{\text{expected appetite}} \\
+ \theta_4 \cdot & \underbrace{\left[-\frac{1}{2}\sum_{j \in \{\mathbf{m},\mathbf{f}\}} \sum_{a_i \in \mathcal{A}_i(h_2^j)} \left(\sigma^i(a_i|h_2^j) - \frac{1}{2}\right)^2\right]}_{\text{expected belief uniformity}} \\
+ \theta_5 \cdot & \underbrace{\left[\sigma^i(\text{App}|h_2^{\mathbf{m}}) \cdot \sigma^i(\text{Ent}|h_2^{\mathbf{m}}) + \sigma^i(\text{App}|h_2^{\mathbf{f}}) \cdot \sigma^i(\text{Ent}|h_2^{\mathbf{f}})\right]}_{\text{expected outcome unlikeliness}} \\
+ \theta_6 \cdot & \underbrace{\frac{1}{2}\left[-\mathbf{1}_{[0,\infty)}(\theta_1) \cdot \sigma^i(\text{App}|h_2^{\mathbf{m}}) - \mathbf{1}_{(-\infty,0]}(\theta_1) \cdot \sigma^i(\text{Ent}|h_2^{\mathbf{m}})\right]}_{\text{expected impatience}} \\
+ \theta_7 \cdot & \underbrace{\theta_1 \cdot \sigma^i(\text{App}|h_2^{\mathbf{m}}) \cdot \sigma^i(\text{App}|h_2^{\mathbf{f}})}_{\text{expected relief / distress}}
\end{aligned}$$

Notice that the "expected surprise" does not appear in the continuation utility because it is always 0. Indeed, surprise cannot be expected: if it were, it would no longer be surprising. Finally, the decision utility of playing App at history $h_2^{\mathbf{m}}$ is:

$$\begin{aligned}
\widehat{U}(\text{App}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 & + \theta_1 + \theta_2 \cdot \mathbf{1}_{\{0\}}(\sigma^i(\text{App}|h_2^{\mathbf{m}})) \\
& + \theta_3 \cdot \frac{1}{2}\left[\mathbf{1}_{[0,\infty)}(\theta_1) \cdot \sigma^i(\text{App}|h_2^{\mathbf{m}}) + \mathbf{1}_{(-\infty,0]}(\theta_1) \cdot \sigma^i(\text{Ent}|h_2^{\mathbf{m}})\right]^2 \\
& + \theta_5 \cdot \sigma^i(\text{Ent}|h_2^{\mathbf{m}}) + \theta_7 \cdot (1 + \theta_1) \cdot \frac{1}{2}\sigma^i(\text{App}|h_2^{\mathbf{f}})
\end{aligned}$$

Other instances of the utility functions can be derived in a similar way. □

3.3. Planning.

We now discuss two criteria to assess the optimality of a plan, and one mode of planning satisfying each criterion. Throughout what follows, we fix an arbitrary decision problem.

Contingent selves. An important feature of our framework is its multi-self approach (see Strotz, 1955). At every history $h \in \widehat{H}$, the agent embodies a different version of himself: we call these "contingent selves." We treat contingent selves as independent entities that reason about each other. For all $h \in \widehat{H}$, we refer to the contingent self who exists (and, if possible, acts) at history h as the h-self. Each contingent self has exclusive control over the action played at the history where he exists. As the agent's preferences change over time, different contingent selves might have contrasting incentives. This is what leads to dynamic inconsistency, discussed in Section 4.3.

Best replies. For an agent with traits $\theta \in \Theta$ and beliefs $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$, the set of best replies at history $h \in H$ is:

$$r_h(\theta, \boldsymbol{\sigma}) := \arg\max_{a_i \in \mathcal{A}_i(h)} \widehat{U}(a_i, h, \theta, \boldsymbol{\sigma})$$

One-step optimality. For an agent with traits $\theta \in \Theta$ and conjecture $\sigma^c \in \Delta^c$, the plan $\sigma^i_* \in \Delta^i$ is one-step optimal if for all $h \in H$:

$$\text{supp } \sigma^i_*(\cdot|h) \subseteq r_h(\theta, \sigma^i_*, \sigma^c)$$

This is a local optimality condition: it constrains the plan component-by-component. It requires the agent to hold a plan that ascribes positive probability only to best replies. In other words, a plan is one-step optimal if, when holding it, the agent has no incentive to deviate from it.

Sequential optimality. For an agent with traits $\theta \in \Theta$ and conjecture $\sigma^c \in \Delta^c$, the plan $\sigma^i_* \in \Delta^i$ is sequentially optimal if for all $h \in \widehat{H}$:

$$\sigma^i_* \in \arg\max_{\sigma^i \in \Delta^i} \vec{U}(h, \theta, \sigma^i, \sigma^c)$$

This is a global optimality condition: it constrains the entire plan simultaneously. It requires the agent to hold a plan that maximizes his continuation utility at every history. In other words, a plan is sequentially optimal if it is (weakly) preferred to all other plans by *every* contingent self.

Rational planning. A rational planner understands that he must accommodate the preferences of his future selves, because they ultimately decide whether to follow the plan or deviate from it. Consequently, he approaches the act of planning as a (folding-back) sequence of local optimizations. We say that the agent plans rationally if he holds a one-step optimal plan.

Unbiased planning. An unbiased planner is not willing to sacrifice the psychological utility of any of his contingent selves to reach an "intertemporal compromise." Consequently, he approaches the act of planning as a sequence of global optimizations. We say that the agent plans unbiasedly if he holds a sequentially optimal plan.

We described sequential optimality as a global requirement, and one-step optimality as a local requirement. This seems to suggest that one implies the other, i.e. all unbiased plans should be rational. This is an intuitive and desirable property, but it does not always hold. Unbiased planning only requires the contingent selves to reach a peaceful agreement, not to reason in a sophisticated way about each

other's incentives. In this sense, unbiased planning is a naïve form of planning. Consequently, some unbiased plans may not be rational. Nonetheless, we adopt the words "global" and "local" because there are ways to restore the desired implication (see Lemma 2.1).

Fix some arbitrary $\theta \in \Theta$, $\sigma^i_* \in \Delta^i$, and $\sigma^c \in \Delta^c$. Table 2 contains the formal definitions of deterministic, unbiased, and rational planning.

The agent plans...	...if σ^i_* belongs to...
deterministically	$\mathrm{DP} := \bigcap_{h\in H}\{\sigma^i \in \Delta^i \colon |\mathrm{supp}\ \sigma^i(\cdot|h)| = 1\}$
unbiasedly	$\mathrm{UP}(\theta,\sigma^c) := \bigcap_{h\in \widehat{H}} \arg\max_{\sigma^i\in\Delta^i} \vec{U}(h,\theta,\sigma^i,\sigma^c)$
rationally	$\mathrm{RP}(\theta,\sigma^c) := \bigcap_{h\in H}\{\sigma^i \in \Delta^i \colon \sigma^i(r_h(\theta,\sigma^i,\sigma^c)|h) = 1\}$

Table 2: Plan-related definitions.

4. Results on spontaneity and self consistency.

We discuss the complications that can arise when the agent's utility depends on his own plan, and the assumptions under which they can be fixed. We proceed in decreasing order of severity.

4.1. Existence.

Under own-plan dependence, a rational agent must construct his own plan not only as a best reply to his conjecture, but also as a best reply *to itself*. The self-referent nature of this observation can be made explicit with a fixed-point argument.

Remark 1.1. *Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. The elements of $\mathrm{RP}(\theta,\sigma^c)$ correspond to the fixed points of the following correspondence:*

$$\begin{aligned}\mathbf{R}^\theta_{\sigma^c} \colon \Delta^i &\rightrightarrows \Delta^i \\ \sigma^i &\mapsto \bigcap_{h\in H}\{\widehat{\sigma}^i \in \Delta^i \colon \widehat{\sigma}^i(r_h(\theta,\sigma^i,\sigma^c)|h) = 1\}\end{aligned}$$

If $\mathbf{R}^\theta_{\sigma^c}$ is not sufficiently well-behaved, it may not have a fixed point. That is, a rational plan may fail to exist entirely. As with most issues that boil down to a matter of continuity, this scenario does not have an enlightening interpretation. Still, we present an example to show exactly how things break down.

Running example (no rational plans).
Let $\theta = (1,2,0,0,0,0,0)$. This means that the agent prefers appetizers over entrées, and likes to be surprised. Specifically, the agent experiences psychological utility when receiving outcomes he previously believed to be impossible. Intuitively then, rationality fails for a very simple reason: the agent would like to surprise himself, but cannot plan to do it (otherwise it would not be a surprise). The only way he can surprise himself is by forming a deterministic plan and then deviating from it. However, if the agent has an incentive to deviate from his plan, the plan cannot be rational. Hence, there exists no rational plan. Here are the utility computations.

$$\widehat{U}(\mathrm{App}, h^{\mathbf{m}}_2, \theta, \boldsymbol{\sigma}) = \vec{U}(h^{\mathbf{ma}}_3, \theta, \boldsymbol{\sigma}) = 2 + 2\cdot \mathbf{1}_{\{0\}}(\sigma^i(\mathrm{App}|h^{\mathbf{m}}_2))$$

$$\widehat{U}(\text{Ent}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{me}}, \theta, \boldsymbol{\sigma}) = 1 + 2 \cdot \mathbf{1}_{\{0\}}(\sigma^i(\text{Ent}|h_2^{\mathbf{m}}))$$
$$r_{h_2^{\mathbf{m}}}(\theta, \boldsymbol{\sigma}) = \begin{cases} \{\text{App}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) < 1 \\ \{\text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) = 1 \end{cases}$$

The last equation is sufficient to conclude that $\text{RP}(\theta, \sigma^c) = \emptyset$. Moreover, the first two equations are sufficient to conclude that $\text{UP}(\theta, \sigma^c) = \emptyset$. Indeed, the first one implies that $\sigma^i \in \text{UP}(\theta, \sigma^c)$ only if $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = 0$, and the second one implies that $\sigma^i \in \text{UP}(\theta, \sigma^c)$ only if $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = 1$. So not only is there no rational plan, there is also no unbiased plan. □

Fortunately, there is a mild assumption we can make to avoid similar scenarios. By assuming that decision utility is continuous in the agent's plan, we can invoke Berge's Theorem to ensure that $\mathbf{R}^{\theta}_{\sigma^c}$ is upper-hemicontinuous, and then use Kakutani's Theorem to prove that a fixed point exists.[20] This assumption is not too strong as it only rules out fringe cases, though some of them may be of interest such as the preference for surprise.

Own plan continuity (OPC). A decision problem satisfies OPC if for all $h \in H$, $a_i \in \mathcal{A}_i(h)$, $\theta \in \Theta$ and $\sigma^c \in \Delta^c$, the decision utility $\widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c)$ is continuous in σ^i.

Notice that OPC cannot be satisfied by the agent alone, or by the decision tree alone: it can only be satisfied by the entire decision problem, i.e. the combination of agent and decision tree. This is because OPC is a property of the decision utility function, which is a hybrid object that combines features of the game (e.g. actions and histories) with features of the agent (e.g. psychological preferences). All properties introduced in Sections 4.1-4.3 follow the same observation. In Section 4.4 we comment on advantages and disadvantages of this approach, and briefly discuss an alternative.

Theorem 1. *Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. Under OPC, there exists a rational plan:* $\text{RP}(\theta, \sigma^c) \neq \emptyset$.

4.2. Spontaneity.

Even though a rational plan surely exists, under OPC alone there might not be a deterministic one. This means that a rational agent may be forced to remain undecided and act spontaneously. This time, the intuition is more insightful. Ideally, a rational planner constructs his plan with a folding-back algorithm (a sort of "intrapersonal backward induction"): he starts by determining the last components of his plan (those corresponding to the longest histories), and then plans backward through the decision tree, down to its root.[21] The folding-back nature of this algorithm reflects the idea that rational planners accommodate the preferences of their future selves.

However, own-plan dependence might break this desirable logic. The algorithm may get stuck at histories where the set of best replies depends on components of the plan that have not yet been determined by the algorithm itself. In other words, early steps of the algorithm depend on the output of later

[20] Essentially, we are following the proof of Nash's Theorem to ensure the existence of an intra-personal equilibrium. For this reason, we omit the proof of Theorem 1.

[21] When two components have the same priority, because they correspond to histories of the same length (as in our running example), the agent can determine them in arbitrary order.

steps, leading to circular reasoning. When this occurs, the only way to identify the rational plan(s) is to abandon the sequential algorithm and proceed simultaneously. In practice, this entails fine-tuning plan components to satisfy the indifference conditions of an intra-personal equilibrium. We illustrate with an example.

Running example (no deterministic rational plans).
Let $\theta = (1, 0, 4, 0, 0, 0, 0)$. This means that the agent prefers appetizers over entrées, and enjoys eating with appetite. Specifically, the more he expects to eat his favorite meal, the stronger his appetite becomes, so the more he enjoys eating any meal. However, his hunger increases with his appetite, and reduces the utility he derives from eating small appetizers instead of full entrées.

The reason why a deterministic rational plan fails to exist is that planning to eat pâté increases the agent's appetite (which he likes), but increases his hunger to the point that eating beef satisfies him more. Conversely, planning to eat beef does not increase his appetite, giving him no reason to avoid the pâté. To solve this dilemma, the agent must remain undecided between the two meals, so whichever he picks maximizes his utility: the pâté because it is his favorite, and the beef because he is sufficiently hungry. Here are the utility computations.

$$\widehat{U}(\text{App}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{ma}}, \theta, \boldsymbol{\sigma}) = 2 + 2\sigma^i(\text{App}|h_2^{\mathbf{m}})^2$$
$$\widehat{U}(\text{Ent}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{me}}, \theta, \boldsymbol{\sigma}) = 1 + 4\sigma^i(\text{App}|h_2^{\mathbf{m}})^2$$
$$\widehat{U}(\text{App}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{fa}}, \theta, \boldsymbol{\sigma}) = \widehat{U}(\text{Ent}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) + 1 = \vec{U}(h_3^{\mathbf{fe}}, \theta, \boldsymbol{\sigma}) + 1 = 0$$
$$\vec{U}(h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 + \sigma^i(\text{App}|h_2^{\mathbf{m}}) + \sigma^i(\text{App}|h_2^{\mathbf{m}})[2 + 2\sigma^i(\text{App}|h_2^{\mathbf{m}})^2] + \sigma^i(\text{Ent}|h_2^{\mathbf{m}})[1 + 4\sigma^i(\text{App}|h_2^{\mathbf{m}})^2]$$
$$\vec{U}(h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = -\sigma^i(\text{Ent}|h_2^{\mathbf{f}}) \qquad \vec{U}(h_1, \theta, \boldsymbol{\sigma}) = \vec{U}(\varnothing, \theta, \boldsymbol{\sigma}) = \frac{\vec{U}(h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) + \vec{U}(h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma})}{2}$$
$$r_{h_2^{\mathbf{f}}}(\theta, \boldsymbol{\sigma}) = \{\text{App}\} \qquad r_{h_2^{\mathbf{m}}}(\theta, \boldsymbol{\sigma}) = \begin{cases} \{\text{App}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) < \sqrt{2}/2 \\ \{\text{App}, \text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) = \sqrt{2}/2 \\ \{\text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) > \sqrt{2}/2 \end{cases}$$

The decision problem satisfies OPC, so a rational plan surely exists. Upon inspection of the utility functions we find that $\sigma^i \in \text{RP}(\theta, \sigma^c)$ iff $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = \sqrt{2}/2$ and $\sigma^i(\text{App}|h_2^{\mathbf{f}}) = 1$. Similarly, $\sigma^i \in \text{UP}(\theta, \sigma^c)$ iff $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = \sigma^i(\text{App}|h_2^{\mathbf{f}}) = 1$.

There is a unique rational plan and a unique unbiased plan, however they do not coincide. The rational plan is not unbiased because, to make himself indifferent between pâté and beef, the agent must lower his appetite decreasing the future enjoyment of either meal. Conversely, every contingent self "agrees" to increase the appetite as much as possible, but doing so creates an incentive (hunger) to deviate from the plan.

In this example, the folding-back algorithm fails immediately. When the algorithm reaches $h_2^{\mathbf{m}}$, the corresponding plan component $\sigma^i(\cdot|h_2^{\mathbf{m}})$ has not yet been determined; however, the set of best replies at $h_2^{\mathbf{m}}$ depends on it. Therefore, the plan component and the associated set of best replies can only be determined simultaneously. □

When a deterministic rational plan fails to exist, the agent himself cannot anticipate exactly what

he is going to do. As a result, it becomes harder for us to predict his behavior. Computing rational plans is also trickier, because we can neither rely on the folding-back algorithm, nor proceed by elimination (since we must now search through the uncountable set of all possible plans). Moreover, the fact that unbiased plans might not be rational challenges the straightforward intuition that global optimality should imply local optimality.

These complications disappear if we assume that the agent's utility is own-plan independent (as is commonly done in the literature). However, this assumption rules out interesting psychological motivations such as anticipatory emotions. Fortunately, to guarantee the existence of a deterministic rational plan (and ensure that every unbiased plan is rational), it suffices to assume "past and counterfactual plan independence:" at every history, the agent's decision utility cannot depend on past or counterfactual components of his plan; however, it may depend on future components.

Past and counterfactual plan independence (PCPI). A decision problem satisfies PCPI if for all $h \in H$, $a_i \in \mathcal{A}_i(h)$, $\theta \in \Theta$, $\sigma^c \in \Delta^c$, and $\widehat{\sigma}^i, \widetilde{\sigma}^i \in \Delta^i$ such that $(\widehat{\sigma}^i(\cdot|h'))_{h' \succ h}^{h' \in H} = (\widetilde{\sigma}^i(\cdot|h'))_{h' \succ h}^{h' \in H}$, we have $\widehat{U}(a_i, h, \theta, \widehat{\sigma}^i, \sigma^c) = \widehat{U}(a_i, h, \theta, \widetilde{\sigma}^i, \sigma^c)$.[22]

PCPI is especially useful because it entails that all rational plans (including a deterministic one) can be found using the folding-back algorithm. PCPI ensures that, at every history reached by the algorithm, the set of best replies only depends on components of the plan that have already been determined. In other words, early steps of the algorithm cannot depend on the output of later steps, because later steps involve histories that are past or counterfactual to those involved in early steps. Simply put, the kind of plan dependence allowed under PCPI is innocuous to the algorithm.

Theorem 2. *Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. Under PCPI, there exists a deterministic rational plan:* $\mathrm{RP}(\theta, \sigma^c) \cap \mathrm{DP} \neq \emptyset$.

The proof of Theorem 2 (Appendix A) contains an explicit construction of the folding-back algorithm. Note that Theorem 2 does not require OPC, which is not implied by PCPI. Another useful consequence of PCPI is that it restores the intuition that sequential optimality implies one-step optimality. In other words, if the agent plans unbiasedly he also plans rationally, though the converse may not hold.[23]

Lemma 2.1. *Under PCPI, for all $\theta \in \Theta$ and $\sigma^c \in \Delta^c$, we have* $\mathrm{UP}(\theta, \sigma^c) \subseteq \mathrm{RP}(\theta, \sigma^c)$.

4.3. (In)consistency.

Under PCPI, all unbiased plans are rational. However, some rational plans might be biased. When this is the case, we say that the agent is self inconsistent. We focus on two interesting types of self inconsistency: plan inconsistency and dynamic inconsistency. These are neither mutually exclusive nor jointly exhaustive; other types of inconsistency exist.

Plan inconsistency. Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. We say that the agent is plan inconsistent if there exist some $\widehat{\sigma}^i, \widetilde{\sigma}^i \in \mathrm{RP}(\theta, \sigma^c)$ such that for all $h \in \widehat{H}$ we have $\vec{U}(h, \theta, \widehat{\sigma}^i, \sigma^c) \leq \vec{U}(h, \theta, \widetilde{\sigma}^i, \sigma^c)$, and

[22] Recall that $(\sigma^i(\cdot|h'))_{h' \succ h}^{h' \in H} = \varnothing$ if there is no $h' \in H$ such that $h' \succ h$ (using Table 1, row 5 column 3).

[23] In view of footnote 8 and Theorem 2, PCPI ensures the existence of a pure personal equilibrium when translated into the framework of Kőszegi (2010).

for some $h' \in \widehat{H}$ we have $\vec{U}(h', \theta, \widehat{\sigma}^i, \sigma^c) < \vec{U}(h', \theta, \widetilde{\sigma}^i, \sigma^c)$. That is, there exists a rational plan which Pareto-dominates another rational plan.[24] Under plan inconsistency, some rational plans are "universally better" than others.

An agent who holds a Pareto-dominated rational plan is still a rational planner, in the sense that he prefers not to deviate from the plan he holds. However, could he switch to a different one, he should do it. In this sense, plan inconsistency is a form of myopia. Some rational agents search for rational plans, but as soon as they find one they stop looking for improvements. Satisfied with their plan, they fail to "see the bigger picture," and realize that other rational plans may yield higher continuation utility.

Running example (plan inconsistency).
Let $\theta = (0, 0, 0, 1, 0, 0, 0)$. This means that the agent is indifferent between appetizers and entrées, and enjoys the suspense of not knowing what he is going to eat. This feeling is stronger the closer his beliefs are to a non-degenerate uniform distribution. Here are the utility computations.

$$\widehat{U}(\text{App}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{ma}}, \theta, \boldsymbol{\sigma}) = \widehat{U}(\text{Ent}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{me}}, \theta, \boldsymbol{\sigma}) = 1$$

$$\widehat{U}(\text{App}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{fa}}, \theta, \boldsymbol{\sigma}) = \widehat{U}(\text{Ent}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{fe}}, \theta, \boldsymbol{\sigma}) = -1$$

$$\vec{U}(h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 = -\vec{U}(h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) \qquad \vec{U}(h_1, \theta, \boldsymbol{\sigma}) = -\frac{1}{2} \sum_{j \in \{\mathbf{m}, \mathbf{f}\}} \sum_{a_i \in \mathcal{A}_i(h_2^j)} \left(\sigma^i(a_i | h_2^j) - \frac{1}{2} \right)^2$$

$$\vec{U}(\varnothing, \theta, \boldsymbol{\sigma}) = \vec{U}(h_1, \theta, \boldsymbol{\sigma}) - \sum_{j \in \{\mathbf{m}, \mathbf{f}\}} \sum_{a_i \in \mathcal{A}_i(h_2^j)} \left(\frac{\sigma^i(a_i | h_2^j)}{2} - \frac{1}{2 + \mathbf{1}_{(0,1)}(\sigma^i(\text{App} | h_2^{\mathbf{m}})) + \mathbf{1}_{(0,1)}(\sigma^i(\text{App} | h_2^{\mathbf{f}}))} \right)^2$$

$$r_{h_2^{\mathbf{f}}}(\theta, \boldsymbol{\sigma}) = r_{h_2^{\mathbf{m}}}(\theta, \boldsymbol{\sigma}) = \{\text{App}, \text{Ent}\}$$

The decision problem satisfies PCPI (although not OPC), so a deterministic rational plan surely exists. Actually, in this example every plan is rational: $\text{RP}(\theta, \sigma^c) = \Delta^i$. However, $\sigma^i \in \text{UP}(\theta, \sigma^c)$ iff $\sigma^i(\text{App} | h_2^{\mathbf{m}}) = \sigma^i(\text{App} | h_2^{\mathbf{f}}) = 1/2$. The only unbiased plan is rational, but mixed.

The agent's $h_2^{\mathbf{m}}$- and $h_2^{\mathbf{f}}$-selves have no psychological motivations, and only care about material satisfaction. Therefore, they are indifferent between meals, and have no incentive to deviate from any plan. In this sense, every plan is rational. However, the agent's h_1- and $\varnothing$-selves strictly prefer to hold a uniform plan. Therefore, the unique unbiased plan Pareto-dominates all other rational plans. □

Now, this is troubling. How useful is it to know that a deterministic rational plan exists if the "best one" might be mixed? We have three possible replies. First, even in standard game theory, a continuum of rational plans can exist alongside two or more deterministic ones. Granted, in standard game theory the agent cannot have a strict incentive to hold a mixed plan. However, this does not guarantee that he will hold a deterministic one, nor does it help us predict which one. Whenever two or more rational plans exist, making behavioral predictions is inherently harder, whether or not the agent's utility is own-plan independent. Second, even if the "best" rational plan is mixed, rationality does not require the

[24] The use of Pareto-dominance makes sense when thinking of contingent selves as separate players indexed by $h \in \widehat{H}$.

agent to hold it.[25] Indeed, holding any rational plan *is* rational, because every rational plan is credibly self-enforcing. If the first two replies do not persuade you, the third one should. Under PCPI, we can invoke a "no ties" argument to ensure the existence of a unique rational plan, which must therefore be deterministic.

Corollary 2.2. *Fix some arbitrary $\theta \in \Theta$, $\sigma^c \in \Delta^c$, and $\sigma^i_* \in \mathrm{RP}(\theta, \sigma^c)$. Under PCPI, σ^i_* induces no ties if and only if it is unique.*

$$\Big[\forall h \in H,\ |r_h(\theta, \sigma^i_*, \sigma^c)| = 1\Big] \iff |\mathrm{RP}(\theta, \sigma^c)| = 1$$

Whenever the rational plan is unique, it cannot be Pareto-dominated by a different one, so the agent is plan consistent. Moreover, under PCPI, the unique rational plan must be deterministic. Therefore, in all decision problems without ties, we can fully leverage Theorem 2 to greatly simplify the analysis: a unique rational plan exists, it is deterministic, and it can be found using the folding-back algorithm. To illustrate how Corollary 2.2 may fail without PCPI, we give an example of a decision problem with multiple rational plans that induce no ties.

Running example (multiple rational plans with no ties).
Let $\theta = (1, 0, 0, 0, -2, 0, 0)$. This means that the agent prefers appetizers over entrées, and likes events to unfold as he expects. As a sort of confirmation bias, the more he anticipates a given outcome, the more he dislikes receiving something different. Here are the utility computations.

$$\widehat{U}(\mathrm{App}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 2 - 2\sigma^i(\mathrm{Ent}|h_2^{\mathbf{m}}) \qquad \widehat{U}(\mathrm{Ent}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 - 2\sigma^i(\mathrm{App}|h_2^{\mathbf{m}})$$
$$\widehat{U}(\mathrm{App}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = -2\sigma^i(\mathrm{Ent}|h_2^{\mathbf{f}}) \qquad \widehat{U}(\mathrm{Ent}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = -1 - 2\sigma^i(\mathrm{App}|h_2^{\mathbf{f}})$$
$$r_{h_2^{\mathbf{m}}}(\theta, \boldsymbol{\sigma}) = \begin{cases} \{\mathrm{Ent}\} & \text{if} \quad \sigma^i(\mathrm{App}|h_2^{\mathbf{m}}) < 1/4 \\ \{\mathrm{App}, \mathrm{Ent}\} & \text{if} \quad \sigma^i(\mathrm{App}|h_2^{\mathbf{m}}) = 1/4 \\ \{\mathrm{App}\} & \text{if} \quad \sigma^i(\mathrm{App}|h_2^{\mathbf{m}}) > 1/4 \end{cases}$$
$$r_{h_2^{\mathbf{f}}}(\theta, \boldsymbol{\sigma}) = \begin{cases} \{\mathrm{Ent}\} & \text{if} \quad \sigma^i(\mathrm{App}|h_2^{\mathbf{f}}) < 1/4 \\ \{\mathrm{App}, \mathrm{Ent}\} & \text{if} \quad \sigma^i(\mathrm{App}|h_2^{\mathbf{f}}) = 1/4 \\ \{\mathrm{App}\} & \text{if} \quad \sigma^i(\mathrm{App}|h_2^{\mathbf{f}}) > 1/4 \end{cases}$$

The decision problem satisfies OPC (but not PCPI), so a rational plan surely exists. Actually, in this example there are 9 rational plans. Indeed, $\sigma^i \in \mathrm{RP}(\theta, \sigma^c)$ iff $\sigma^i(\mathrm{App}|h_2^{\mathbf{m}}) \in \{0, 1/4, 1\} \ni \sigma^i(\mathrm{App}|h_2^{\mathbf{f}})$. However, none of the deterministic rational plans induces a tie. This happens because the agent's confirmation bias makes him "stubborn." If for some reason he plans to have an entrée, even though he prefers appetizers, it is actually (strictly) better for him to have the entrée. This way, he avoids the psychological disutility of being wrong. □

Dynamic inconsistency. Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. We say that the agent is dynamically inconsistent if $\mathrm{UP}(\theta, \sigma^c) = \emptyset$. The cause of the unbiased planner is noble, but often doomed: sometimes it is simply impossible to satisfy every contingent self. Unbiased plans fail to exist when there is a fundamental incompatibility between the solutions of global optimizations carried out at

[25] Expressed in our language, Kőszegi and Rabin's (2006) notion of preferred personal equilibrium (a refinement of personal equilibrium) requires the agent to hold the "best" rational plan.

different histories. In other words, the agent cannot find a single plan that reconciles the conflicting interests of all contingent selves.

Running example (dynamic inconsistency).
Let $\theta = (1, 0, 0, 0, 0, 2, 0)$. This means that the agent prefers appetizers over entrées, and gets impatient about eating his favorite meal. He cannot wait to have pâté: the more he expects to eat it, the more he aches in anticipation. Here are the utility computations.

$$\widehat{U}(\text{App}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{ma}}, \theta, \boldsymbol{\sigma}) = \widehat{U}(\text{Ent}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) + 1 = \vec{U}(h_3^{\mathbf{me}}, \theta, \boldsymbol{\sigma}) + 1 = 2$$
$$\widehat{U}(\text{App}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = \vec{U}(h_3^{\mathbf{fa}}, \theta, \boldsymbol{\sigma}) = \widehat{U}(\text{Ent}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) + 1 = \vec{U}(h_3^{\mathbf{fe}}, \theta, \boldsymbol{\sigma}) + 1 = 0$$
$$\vec{U}(h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 + \sigma^i(\text{App}|h_2^{\mathbf{m}}) \qquad \vec{U}(h_1, \theta, \boldsymbol{\sigma}) = \frac{1 - \sigma^i(\text{App}|h_2^{\mathbf{m}}) - \sigma^i(\text{Ent}|h_2^{\mathbf{f}})}{2}$$
$$r_{h_2^{\mathbf{f}}}(\theta, \boldsymbol{\sigma}) = r_{h_2^{\mathbf{m}}}(\theta, \boldsymbol{\sigma}) = \{\text{App}\}$$

The decision problem satisfies PCPI, so a deterministic rational plan surely exists. Indeed, $\sigma^i \in \text{RP}(\theta, \sigma^c)$ iff $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = \sigma^i(\text{App}|h_2^{\mathbf{f}}) = 1$. However, $\text{UP}(\theta, \sigma^c) = \emptyset$. Indeed, the second-to-last equation implies that $\sigma^i \in \text{UP}(\theta, \sigma^c)$ only if $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = 0$, and the third-to-last equation implies that $\sigma^i \in \text{UP}(\theta, \sigma^c)$ only if $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = 1$.

The agent's $h_2^{\mathbf{m}}$- and $h_2^{\mathbf{f}}$-selves have no psychological motivations, and only care about material satisfaction. Therefore, they only follow plans that prescribe the best feasible meal (i.e. an appetizer). Understanding this, a rational planner has no choice but to plan accordingly. However, the agent's h_1-self dislikes the resulting rational plan, because it makes him feel impatient: this is the unbiased planner's dilemma. Since our framework precludes any form of self-deception, there is simply no way to solve it. □

Self consistency. Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. We say that the agent is self consistent if $\text{RP}(\theta, \sigma^c) = \text{UP}(\theta, \sigma^c) \neq \emptyset$. Clearly, if the agent is self consistent he is also plan consistent and dynamically consistent.

Self inconsistency is not particularly problematic, as it alone does not lead to a failure of the folding-back algorithm. Still, it may be helpful to know a property that ensures consistency.

Own-plan independence (OPI). The decision problem satisfies OPI if for all $t \in \mathbb{T}$, $z \in Z$, $\theta \in \Theta$, $\sigma^c \in \Delta^c$, and $\widehat{\sigma}^i, \widetilde{\sigma}^i \in \Delta^i$, we have $U_t(z, \theta, \widehat{\sigma}^i, \sigma^c) = U_t(z, \theta, \widetilde{\sigma}^i, \sigma^c)$.

Remark 3.1. *OPI implies both PCPI and OPC.*

OPI is the strongest property. As such, it yields the strongest result.

Theorem 3. *Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. Under OPI, the following hold:*

1. *There exists a deterministic rational plan:* $\text{RP}(\theta, \sigma^c) \cap \text{DP} \neq \emptyset$.
2. *The agent is self consistent:* $\text{RP}(\theta, \sigma^c) = \text{UP}(\theta, \sigma^c) \neq \emptyset$.

The first point of Theorem 3 is well-known in the literature, and a corollary to Theorem 2. The second point is essentially a folk result. We omit the proof, as it is a bookkeeping exercise relying on standard methods of dynamic programming. As a consequence of the second point, the one-deviation principle holds: sequential optimality and one-step optimality are equivalent.

Corollary 3.2. *Fix some arbitrary* $h \in \widehat{H}$, $\theta \in \Theta$, $\sigma^c \in \Delta^c$, *and* $\widehat{\sigma}^i, \widetilde{\sigma}^i \in \mathrm{RP}(\theta, \sigma^c)$. *Under OPI, we have* $\vec{U}(h, \theta, \widehat{\sigma}^i, \sigma^c) = \vec{U}(h, \theta, \widetilde{\sigma}^i, \sigma^c)$.

Under OPI, all rational plans yield the same continuation utility, so no rational plan can Pareto-dominate the others: this ensures plan consistency. Another consequence of Corollary 3.2 is that under OPI the set of rational plans must be convex. Without OPI this is not always the case, even if PCPI holds.

OPI is a stringent requirement that rules out many interesting psychological motivations, but it can be appropriate in the right context. In games of incomplete information, the agent's utility may depend exclusively on his beliefs about unknown parameters, such as his own intelligence. In multi-player games, the agent's utility may depend exclusively on his co-players' beliefs, as in Battigalli and Dufwenberg (2007). Even in simple decision problems, the structure of the decision tree may be such that the agent's utility depends exclusively on his conjecture about chance. These are all examples of psychological games where the agent has a belief-dependent utility function, but OPI holds.

4.4. Concluding remarks.

A few remarks are in order. First, we can strengthen the connection with the existing literature by organizing common classes of psychological games according to the properties they satisfy. Games with regret and rejoice (Battigalli et al., 2025), reciprocity (Dufwenberg and Kirchsteiger, 2004), and simple guilt (Battigalli and Dufwenberg, 2007) all satisfy OPI.[26] Games with anticipatory emotions, such as anxiety and suspense (Caplin and Leahy, 2001) or savoring and dread (Loewenstein, 1987), do not satisfy OPI but do satisfy PCPI. Finally, games with disappointment and elation, also known as belief-dependent reference-dependence (Kőszegi and Rabin, 2006), and games with frustration and anger (Battigalli et al., 2019b), generally fail to satisfy PCPI and satisfy only OPC.

Since our framework only allows for a single personal player, the word "own" in OPC and OPI is redundant, as there is no other plan on which the agent's utility may depend. However, in the multi-player generalization of our framework, each player's utility may well depend on his co-players' plans, and this dependence may not be continuous. The good news is that, for the purposes of spontaneity, rationality and self-consistency, we simply do not care about dependence on other players' plans. As long as OPC, PCPI, or OPI are satisfied exactly as stated in this paper, Theorems 1, 2 or 3 continue to hold. This is because each player only chooses his own plan, so all other types of behavioral beliefs (including his co-players' plans) are just parameters in his planning process.

As mentioned before, we defined OPC, PCPI, and OPI in terms of hybrid utility functions (decision and path utility). The advantage of this approach is that it applies more generally, including cases where the agent's preferences depend on features of the decision tree (something we did not allow in this paper). The drawback is that we cannot check whether each property holds based only on the agent's preferences; we must also know the specific decision tree he is facing. It would be much more convenient if we could determine, just by looking at the agent's preferences, that OPC, PCPI or OPI are satisfied in *every* decision problem he may face. Fortunately, this can be achieved quite easily.

[26] We use the term simple guilt in contrast to guilt from premeditation, which appears in our first heuristic example. That example fails to satisfy PCPI, let alone OPI.

Own-belief continuity (OBC). The agent satisfies OBC if for all $t \in \mathbb{T}$, $y^t \in Y^t$, and $\theta \in \Theta$, the instantaneous utility $\nu_t(y^t, \theta, \mu_0^t)$ is continuous in μ_0^t.

Past belief independence (PBI). The agent satisfies PBI if for all $t \in \mathbb{T}$, $y^t \in Y^t$, $\theta \in \Theta$, and $\widehat{\mu}_0^t, \widetilde{\mu}_0^t \in M^{t+1}$ such that $\widehat{\mu}_t = \widetilde{\mu}_t$, we have $\nu_t(y^t, \theta, \widehat{\mu}_0^t) = \nu_t(y^t, \theta, \widetilde{\mu}_0^t)$.

Own-belief independence (OBI). The agent satisfies OBI if for all $t \in \mathbb{T}$, $y^t \in Y^t$, $\theta \in \Theta$, and $\widehat{\mu}_0^t, \widetilde{\mu}_0^t \in M^{t+1}$, we have $\nu_t(y^t, \theta, \widehat{\mu}_0^t) = \nu_t(y^t, \theta, \widetilde{\mu}_0^t)$.

Remark 3.3. *If the agent satisfies OBC (resp. PBI, resp. OBI), any decision problem he faces satisfies OPC (resp. PCPI, resp. OPI).*

Conversely, the fact that a decision problem satisfies OPC, PCPI or OPI does not imply that the agent satisfies OBC, PBI or OBI. With a single personal player, OBI actually implies total belief independence, so it effectively reduces the analysis to standard game theory. However, in the multi-player generalization of our framework, OBI can hold even if the agent has some psychological motivation that depends on his co-players' beliefs.

The fact that PBI "implies" PCPI may seem puzzling. Should the latter not be stronger, since it refers to both past and counterfactual beliefs? The difference lies in the type of beliefs that the two properties involve. PBI requires independence from past material beliefs, whereas PCPI requires independence from past and counterfactual beliefs about own behavior. As it turns out, counterfactual plans can be mapped into past material beliefs: if the agent's decision utility depends on a counterfactual component of his plan, then his instantaneous utility must depend on some past material belief, held at a history preceding both the current and the counterfactual one. Relatedly, one may wonder why we need the "counterfactual" part of PCPI, and whether it would suffice to just assume "past plan independence."

Past plan independence (PPI). A decision problem satisfies PPI if for all $h \in H$, $a_i \in \mathcal{A}_i(h)$, $\theta \in \Theta$, $\sigma^c \in \Delta^c$, and $\widehat{\sigma}^i, \widetilde{\sigma}^i \in \Delta^i$ such that $(\widehat{\sigma}^i(\cdot|h'))_{h' \not\geq h}^{h' \in H} = (\widetilde{\sigma}^i(\cdot|h'))_{h' \not\geq h}^{h' \in H}$, we have $\widehat{U}(a_i, h, \theta, \widehat{\sigma}^i, \sigma^c) = \widehat{U}(a_i, h, \theta, \widetilde{\sigma}^i, \sigma^c)$.

Clearly, PCPI implies PPI. To show that they are not equivalent, we must find an example where the agent's decision utility depends on counterfactual components of his plan, without also depending on past components. If we allowed preferences to depend on features of the decision tree, such examples would be easy to find, justifying the use of PCPI. However, in our framework this is not allowed, and examples are harder to find; indeed, they are actually impossible if chance is permanently inactive.

Proposition 3.4. *If $|\Delta^c| = 1$, PPI implies PCPI.*

This explains why we included chance in our framework. Without a second player (personal or not), we cannot fully appreciate the role of PCPI. It is only when chance is active that PPI is no longer sufficient: a decision problem may satisfy PPI without also satisfying PCPI, and a deterministic rational plan may fail to exist. Here is an example.

Running example (insufficiency of PPI).
Let $\theta = (0, 0, 0, 0, 0, 0, 2)$. This means that the agent is indifferent between appetizers and entrées, and

is affected by relief and distress. The more he initially (at date 0) expects to eat an appetizer he dislikes (resp. likes), the more he is relieved (resp. distressed) by eating an appetizer he likes (resp. dislikes). The same logic applies to entrées. Here are the utility computations.

$$\widehat{U}(\text{App}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 + \sigma^i(\text{App}|h_2^{\mathbf{f}}) \qquad \widehat{U}(\text{App}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = -1 - \sigma^i(\text{App}|h_2^{\mathbf{m}})$$
$$\widehat{U}(\text{Ent}, h_2^{\mathbf{m}}, \theta, \boldsymbol{\sigma}) = 1 + \sigma^i(\text{Ent}|h_2^{\mathbf{f}}) \qquad \widehat{U}(\text{Ent}, h_2^{\mathbf{f}}, \theta, \boldsymbol{\sigma}) = -1 - \sigma^i(\text{Ent}|h_2^{\mathbf{m}})$$

$$r_{h_2^{\mathbf{m}}}(\theta, \boldsymbol{\sigma}) = \begin{cases} \{\text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{f}}) < 1/2 \\ \{\text{App}, \text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{f}}) = 1/2 \\ \{\text{App}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{f}}) > 1/2 \end{cases}$$

$$r_{h_2^{\mathbf{f}}}(\theta, \boldsymbol{\sigma}) = \begin{cases} \{\text{App}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) < 1/2 \\ \{\text{App}, \text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) = 1/2 \\ \{\text{Ent}\} & \text{if} \quad \sigma^i(\text{App}|h_2^{\mathbf{m}}) > 1/2 \end{cases}$$

The decision problem satisfies PPI, but not PCPI. It also satisfies OPC, so a rational plan surely exists. However, the rational plan is not deterministic. Indeed, $\sigma^i \in \text{RP}(\theta, \sigma^c)$ iff $\sigma^i(\text{App}|h_2^{\mathbf{m}}) = \sigma^i(\text{App}|h_2^{\mathbf{f}}) = 1/2$.

A deterministic rational plan fails to exist because the agent faces circular incentives. If he plans to have an appetizer (resp. entrée) at the fish restaurant, he should also plan to have an appetizer (resp. entrée) at the meat restaurant, to increase his feeling of relief. However, if he plans to have an appetizer (resp. entrée) at the meat restaurant, he should plan to have an entrée (resp. appetizer) at the fish restaurant, to decrease his feeling of distress. □

This concludes the pure-theory part of our paper. We now turn to applying some of our results to a specific problem of intertemporal choice with savoring and dread.

5. Anticipatory emotions, savoring, and dread.

Anticipatory emotions are those elicited in anticipation of future events. They are the most prominent example of psychological motivations satisfying PBI (so PCPI always holds) but not OBI (so OPI may fail). When studying these emotions, one should be careful not to confuse anticipatory utility (i.e. the utility derived from anticipatory emotions) with expected utility. Anticipatory utility is experienced in the present, and captures the agent's preferences over his current beliefs about the future. Expected utility is the present discounted value of utility that will be experienced in the future.[27] Since future utility is uncertain, the agent computes its expectation using the same beliefs that yield anticipatory utility; this may puzzle the inattentive reader.

5.1. Savoring and dread.

Savoring is the utility derived by anticipating future positive (i.e. pleasant) experiences; dread is the disutility derived by anticipating future negative (i.e. unpleasant) experiences. Loewenstein (1987) first developed a formal theory of savoring and dread. His definitions were not explicitly belief-dependent; here we present a belief-dependent reinterpretation of his model, within the framework of psychological

[27] In the words of Elster and Loewenstein (1992), expected utility can be interpreted as a form of sympathy for one's own future self, whereas anticipatory utility can be interpreted as a form of empathy.

game theory (as detailed in Section 3).

Material utility. Let $m\colon Y \cup \{\aleph\} \to \mathbb{R}$ be the material utility function of the agent. It represents the agent's preferences over the physical experience of material outcomes (recall that we think of $\aleph$ as a null outcome). It is normalized so that its sign indicates whether the agent likes or dislikes an experience, allowing us to distinguish between positive, negative, and null or neutral outcomes. Naturally, we set $m(\aleph) = 0$.

Extension functions. For all $t \in \mathbb{N}$, let $\mathcal{Y}_t\colon Y^{\leq T} \to Y \cup \{\aleph\}$ be an extension function, defined as:

$$\mathcal{Y}_t(y^k) := \begin{cases} y_t & \text{if } \ t \leq k \\ \aleph & \text{if } \ t > k \end{cases}$$

The feasible streams of material outcomes are all finite, but may differ in length. For mathematical convenience, we extend all of these streams with an infinite sequence of null outcomes. This extension does not distort the agent's preferences in any way. From the agent's perspective, it makes no difference whether we model him as receiving a finite stream of outcomes, or an infinite stream of eventually null outcomes.

Vividness. Let $\Phi\colon Y \cup \{\aleph\} \to [0,1]$ be a function that represents how vivid the agent's mental image is of any outcome $y \in Y \cup \{\aleph\}$. The more vividly an outcome is visualized, the more it captures the agent's anticipation. We treat Φ as one of the agent's traits.

Savoring. Let $\mathcal{S}_t^\tau(\mu_t, \Phi)$ be the savoring experienced by the agent at date $t \in \mathbb{T}$ in anticipation of date $\tau \in \{t+1, ..., T\}$, given his belief $\mu_t \in M$ and vividness $\Phi \in [0,1]^{Y \cup \{\aleph\}}$. Formally:

$$\mathcal{S}_t^\tau(\mu_t, \Phi) := \left[\mathbb{E}[(\Phi(\mathcal{Y}_\tau))^{\tau - t} \cdot m(\mathcal{Y}_\tau); \mu_t]\right]^+$$

Let $\mathcal{S}_t(\mu_t, \Phi)$ be the total savoring experienced by the agent at date $t \in \mathbb{T}$, given his belief $\mu_t \in M$ and vividness $\Phi \in [0,1]^{Y \cup \{\aleph\}}$. Formally:

$$\mathcal{S}_t(\mu_t, \Phi) := \begin{cases} \sum_{\tau=t+1}^{T} \mathcal{S}_t^\tau(\mu_t, \Phi) & \text{if } \ t < T \\ 0 & \text{if } \ t = T \end{cases}$$

Dread. Let $\mathcal{D}_t^\tau(\mu_t, \Phi)$ be the dread experienced by the agent at date $t \in \mathbb{T}$ in anticipation of date $\tau \in \{t+1, ..., T\}$, given his belief $\mu_t \in M$ and vividness $\Phi \in [0,1]^{Y \cup \{\aleph\}}$. Formally:

$$\mathcal{D}_t^\tau(\mu_t, \Phi) := \left[\mathbb{E}[(\Phi(\mathcal{Y}_\tau))^{\tau - t} \cdot m(\mathcal{Y}_\tau); \mu_t]\right]^-$$

Let $\mathcal{D}_t(\mu_t, \Phi)$ be the total dread experienced by the agent at date $t \in \mathbb{T}$, given his belief $\mu_t \in M$ and vividness $\Phi \in [0,1]^{Y \cup \{\aleph\}}$. Formally:

$$\mathcal{D}_t(\mu_t, \Phi) := \begin{cases} \sum_{\tau=t+1}^{T} \mathcal{D}_t^\tau(\mu_t, \Phi) & \text{if } \ t < T \\ 0 & \text{if } \ t = T \end{cases}$$

Notice that, by construction, $\mathcal{D}_t^\tau(\mu_t, \Phi) \cdot \mathcal{S}_t^\tau(\mu_t, \Phi) = 0$ for all $t \in \mathbb{T}$, $\mu_t \in M$, $\Phi \in [0,1]^{Y \cup \{\aleph\}}$, and $\tau \in \{t+1, ..., T\}$. In other words, the agent cannot simultaneously experience savoring and dread in

anticipation of the same date τ.

Raising $\Phi(\cdot)$ to the power of $\tau - t$ dampens the anticipation of dates farther in the future. The vividness of future dates decreases with their distance from the present, and so does the intensity of the emotions they elicit. In other words, the greatest savoring (resp. dread) is enjoyed (resp. suffered) right before the expected date of experience. Story et al. (2013) provide empirical evidence for the exponential shape of this effect. The idea of interpreting vividness as a discount factor traces back to Ramsey (1928), who argues that discounting future periods is "a practice which [...] arises merely from the weakness of the imagination." Even earlier, Böhm-Bawerk (1889) cited "the fragmentary nature of the imaginary picture that we construct of the future state of our wants" as a reason why we "systematically undervalue our future wants."

Utility functions. Let $(\theta_{\mathcal{S}}, \theta_{\mathcal{D}}, \Phi, \delta) = \theta \in \Theta = \mathbb{R}_+ \times \mathbb{R}_+ \times [0,1]^{Y \cup \{\aleph\}} \times [0,1]$. The parameters $\theta_{\mathcal{S}}$ and $\theta_{\mathcal{D}}$ capture the agent's sensitivity to savoring and dread, while δ is the intertemporal discount factor. For all $t \in \mathbb{T}$, we specify the period-t instantaneous utility function as:

$$\nu_t(y^t, \theta, \mu_0^t) := m(y_t) + \theta_{\mathcal{S}} \cdot \mathcal{S}_t(\mu_t, \Phi) + \theta_{\mathcal{D}} \cdot \mathcal{D}_t(\mu_t, \Phi) \qquad (\star)$$

Given a decision tree, the intertemporal, path, decision, and continuation utility functions can be derived from ν_t as explained in Section 3.2.

Remark 4.1. *An agent with utility function ($\star$) satisfies PBI.*

5.2. Literature review.

The most prominent behavioral implications of savoring and dread arise in problems of intertemporal choice. To avoid feeling dread, the agent has an incentive to "rip the band-aid off," and hasten the experience of negative outcomes. Conversely, to savor the future, he has an incentive to indulge in anticipation, and delay the experience of positive outcomes. For example, a hospital patient may reschedule a painful operation to an earlier date, and a wine enthusiast may repeatedly delay opening an expensive bottle. These incentives oppose the standard predictions of discounted utility theory, according to which rational agents should prefer to experience immediately (resp. maximally delay) all positive (resp. negative) outcomes. Consequently, Loewenstein (1987) argues that neglecting the role of anticipatory emotions might have led to a systematic underestimation of intertemporal discount rates.

Loewenstein (1987) studies the effects of savoring and dread on the optimal choice of delay under commitment. In the case of dread he finds that, depending on individual parameters, rational agents might prefer immediate experience or maximum delay, but never an intermediate option. We refer to this pattern as "choice polarization." In the case of savoring choice polarization does not occur, as some agents may prefer an intermediate date of experience. In the conclusion of his paper, Loewenstein (1987) mentions that savoring and dread might generate dynamic inconsistency in decision problems under discretion; however, he does not discuss how dynamic inconsistency would affect the strategic reasoning of a sophisticated agent, nor does he characterize its behavioral implications. As the main contribution of the next section, we take on this task.

In the experimental literature, savoring and dread have been studied extensively. Overall, the effects of dread on choice under commitment are supported by strong evidence. The central contribution is by

Berns et al. (2006), who document a strong preference for the immediate experience of an inevitable electric shock; many subjects are even willing to endure a greater shock in order to avoid the dread of waiting. Similar observations appear in older studies (Cook and Barnes Jr, 1964; Hare, 1966a, 1966b; Badia et al., 1966; Belanger and Sattler, 1967) as well as in more recent work (Harris, 2012; Story et al., 2013; Hardisty and Weber, 2020; Story et al., 2020, 2021).

In the case of savoring, the evidence is less decisive. The strongest result is by Di Cagno et al. (2019), who report that 47% of their sample preferred to delay the consumption of a free snack by two or more weeks. However, other studies document that subjects are often statistically indifferent between immediate experience and marginal delay of positive outcomes (Mischel et al., 1969; Lovallo and Kahneman, 2000; Hardisty and Weber, 2020).

With the exception of Di Cagno et al. (2019), all of the experiments mentioned so far were conducted under commitment, either without a theoretical model or within the framework of Loewenstein's model. At the start of the experiment, participants chose when to experience a known outcome, and their initial choice was enforced by the experimenters. As we extend the analysis to choice under discretion, we hope to pave the way for a new line of experimental work.

Compared to previous theories of savoring and dread, we allow for the heterogeneous vividness of different outcomes. This does not play a direct role in the results of Section 6, but it does have an interesting consequence in some problems of choice under uncertainty. When some negative (resp. positive) outcomes have a disproportionately high vividness, the agent may experience dread (resp. savoring) in anticipation of positive (resp. negative) lotteries. In these scenarios, the feeling of dread (resp. savoring) may be referred to as fear (resp. hope). Similar situations are commonly observed.

Heuristic example (fear of flights).
Picture an agent boarding a flight. The plane takes off at date 1, and can either crash or land safely at date 2. The set of outcomes is $Y = \{\text{Crash}, \text{Safe}\}$. Naturally, $m(\text{Crash}) < 0 < m(\text{Safe})$, and $|m(\text{Crash})| > m(\text{Safe})$. Neither outcome is initially ruled out (i.e. μ_1 has full support), but the agent is fairly confident in the safety of the aircraft. The flight is, therefore, a positive lottery: $\mathbb{E}[m(\mathcal{Y}_2); \mu_1] > 0$. Although a crash is extremely unlikely, the agent is very scared of it, and visualizes it much more vividly than a safe landing ($\Phi(\text{Safe}) << \Phi(\text{Crash})$). Assume for simplicity that $\Phi(\text{Safe}) = 0$. Now, we can easily check that $\mathcal{S}_1^2(\mu_1, \Phi) = 0 > \mathcal{D}_1^2(\mu_1, \Phi)$. The agent actively dreads the flight, even though he believes that it will likely be safe.

6. Commitment, discretion, and dynamic inconsistency.

This section contains our applied-theory results. We study a problem of intertemporal choice under commitment and discretion, and provide a (partial) behavioral characterization of dynamic consistency for this class of decision problems. As noted in the introduction, part of our motivation for introducing PCPI and studying its implications was to facilitate the analysis of savoring and dread in decision problems under *discretion*. In fact, Theorem 2 is not needed to solve the commitment problem: even though the instantaneous utility is belief-dependent, the game with commitment is so simple that it

induces a plan-independent decision utility (i.e. it satisfies OPI).[28] By contrast, PCPI and Theorem 2 are essential to solve the game under discretion, which does *not* satisfy OPI.

To emphasize how fundamentally the behavioral implications of savoring and dread differ from the standard predictions of discounted utility theory, we deliberately study the simplest possible decision problem. There is a single deterministic outcome, it is known by the agent, and it must be experienced within a given deadline; the agent can only choose when to experience it. In particular, chance plays no role in this decision problem.[29] Still, all of our results would continue to hold if we replaced the deterministic outcome with a (subjective) lottery over equally vivid outcomes, modeled as a chance move.[30]

All throughout, we assume that the agent's instantaneous utility is specified as in $(\star)$. We compare two versions of the decision problem, one under commitment and one under discretion. Even though the instantaneous utility is the same in both versions, the decision utilities will differ, because the decision trees do not coincide. To streamline the exposition, we omit their derivation from this section, but all the details can be found in Appendix B. Deriving the decision utility under commitment is straightforward, as it is plan-independent (see proof of Propositions 5.1-5.2).

Conversely, the game with discretion induces a plan-dependent decision utility. Deriving this utility function is technically demanding, especially for mixed plans. Fortunately, Theorem 2 and Corollary 2.2 allow us to focus on deterministic plans, greatly simplifying the analysis. This perfectly illustrates the usefulness of our results from Section 4. Even so, deriving the decision utility is not immediate and requires some care (see Lemma 8.2).

6.1. Choice polarization.

The benchmark of our analysis is choice under commitment: the agent must choose when to experience a known outcome, and his initial decision is enforced by the rules of the game. As a sanity check for our model, we begin by replicating Loewenstein's results about choice polarization (or lack thereof): this confirms that our belief-dependent reinterpretation is consistent with the original theory.

Figure 4 depicts a T-periods delay game with commitment, $G_K(T)$ (where K denotes commitment). Chance is permanently inactive, so $\mathcal{A}_c(h) = \{w\}$ for all $h \in H$. The set of material outcomes is $Y = \{x, \aleph\}$. At the initial history, the agent must choose when to experience x, so $\mathcal{A}_i(\varnothing) = \{1, ..., T\} = \mathbb{T}$. Thereafter, the agent becomes inactive and cannot "change his mind." Formally, $\mathcal{A}_i(h) = \{w\}$ for all $h \in H \setminus \{\varnothing\}$. This assumption is justified, for instance, if the outcome is administered by an experimenter or a mechanical device. Throughout all the waiting periods, the agent receives nothing. To

[28]The reason is that the agent's beliefs are pinned down by his initial choice. As a result, Theorem 3 applies to the game with commitment.

[29]While chance played an important role in the general framework developed in the first part of the paper, the purpose of this section is to identify the behavioral implications of savoring and dread. Showing that departures from standard utility theory can arise even without chance demonstrates that they are not driven by stochastic features of the environment.

[30]In the presence of chance, however, one might also want to introduce anticipatory emotions that depend directly on the uncertainty of the lottery, such as anxiety (see Caplin and Leahy, 2001). Anxiety can be defined as aversion to the variance of future outcomes. As an anticipatory emotion, it satisfies PBI, ensuring the existence of a deterministic rational plan. However, it may generate incentives that oppose those of savoring and dread, thereby confounding the analysis.

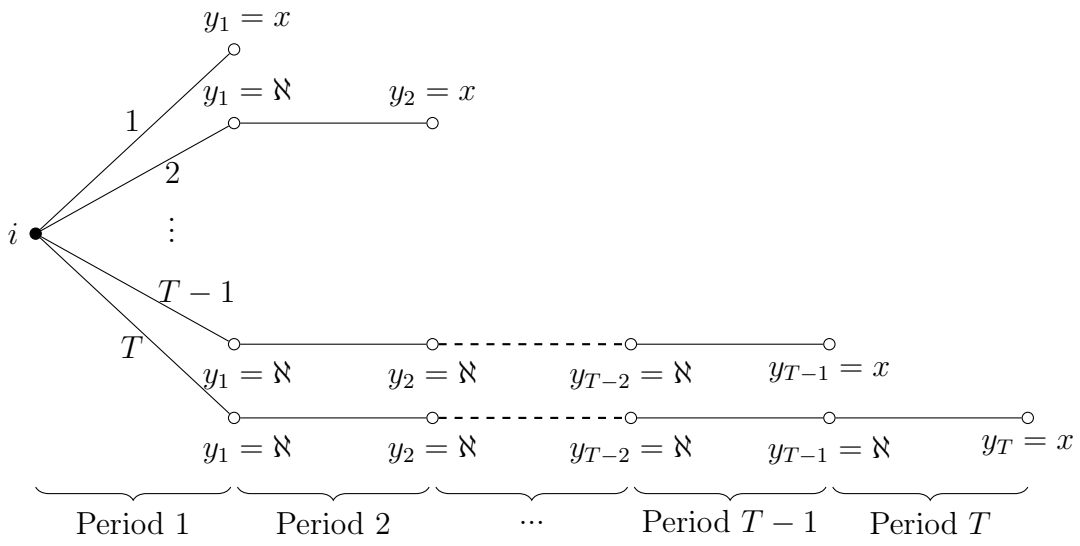


Figure 4: T-periods delay game with commitment $G_K(T)$.

simplify notation, let $m(x) = q$ and $\Phi(x) = \phi$. See the proof of Propositions 5.1-5.2 for a derivation of the decision utility induced by $(\star)$ in $G_K(T)$.

Proposition 5.1. *Fix some arbitrary $T \in \mathbb{N}$, $\theta \in \text{int}(\Theta)$, $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$, and $q < 0$. The agent prefers immediate experience or maximum delay.*

$$r_{\varnothing}(\theta, \boldsymbol{\sigma}) \subseteq \{1, T\}$$

In the case of dread, the agent surely makes an extreme choice: immediate experience, or maximum delay. Depending on θ and T, he may prefer one or the other, but never an intermediate option. The intuition is that delaying the experience has one cost (adding more periods of dread) but two benefits (discounting the present value of x *and* the present value of the greatest dread, which immediately precedes the experience). With a high sensitivity to dread the cost may be higher, but with a low sensitivity the benefits prevail (see Proposition 6.1 for comparative statics on $\theta_{\mathcal{D}}$). Therefore, if the agent is willing to delay the experience by at least one period, he must be willing to delay it all the way. Formally, the linear interpolation of the agent's decision utility is a quasi-convex function, so its maximization must have a boundary solution.

Proposition 5.2. *Fix some arbitrary $T \in \mathbb{N}$, $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$, and $q > 0$. The agent may strictly prefer any feasible amount of delay.*

$$\forall t \in \mathbb{T},\ \exists \theta \in \text{int}(\Theta) \colon r_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{t\}$$

In the case of savoring, different values of θ can make the agent prefer any amount of delay. The intuition is that delaying the experience has one benefit (adding more periods of savoring) but two costs (discounting the present value of x *and* the present value of the greatest savoring, which immediately precedes the experience). With a high sensitivity to savoring the benefit may be higher, but with a low sensitivity the costs prevail (see Proposition 6.2 for comparative statics on $\theta_{\mathcal{S}}$). Therefore, the agent may be willing to delay the experience by some periods, but not necessarily all the way. Formally, the linear interpolation of the agent's decision utility is a quasi-concave function, so its maximization might have an interior solution.

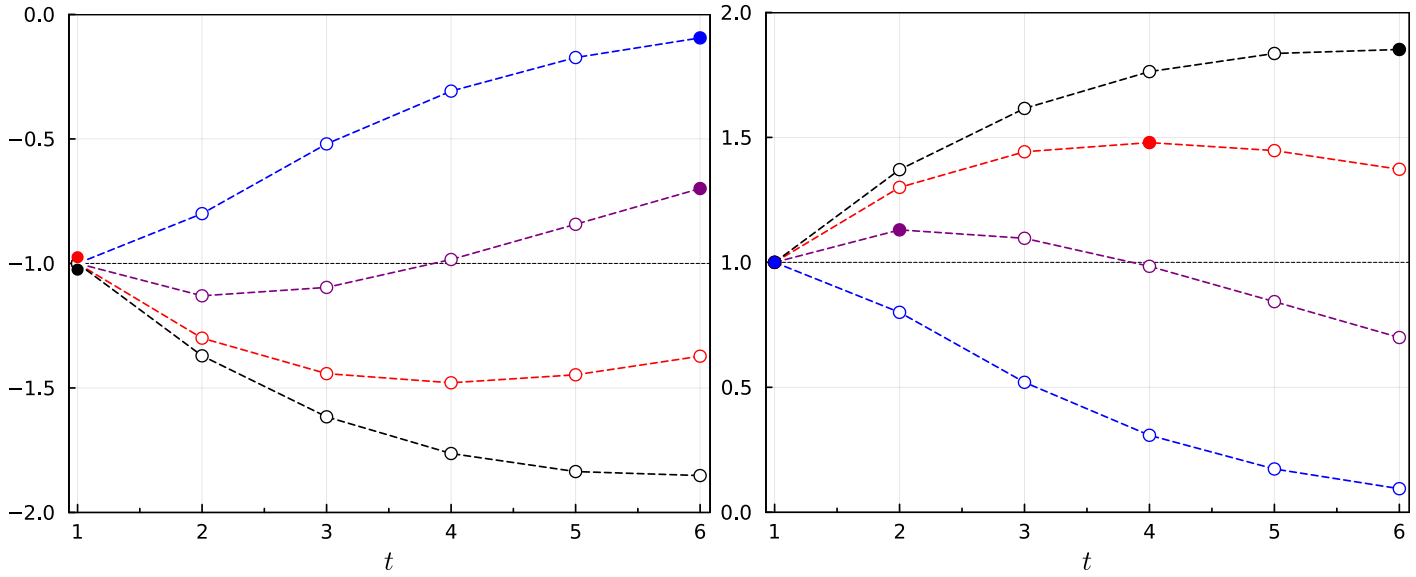


Figure 5: Visualization of $\widehat{U}(t, \varnothing, \theta, \boldsymbol{\sigma})$ for $T = 6$. On the left, $q = -1$. On the right, $q = 1$. Different colors correspond to different values of θ. The filled-in dots mark the best reply for each value of θ.

6.2. Dynamic (in)consistency.

In many real-world scenarios, commitment devices are not available: agents cannot enforce their initial preferences, so they play under discretion.

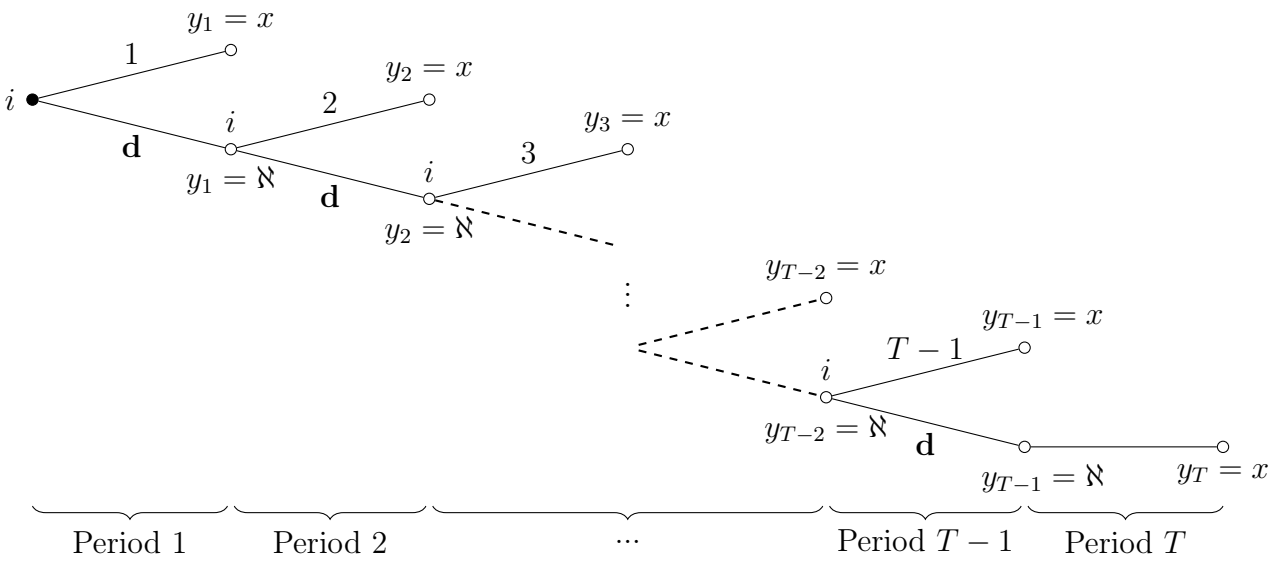


Figure 6: T-periods delay game with discretion $G(T)$.

Figure 6 depicts a T-periods delay game with discretion, $G(T)$. Chance is permanently inactive, so $\mathcal{A}_c(h) = \{w\}$ for all $h \in H$. The set of material outcomes is $Y = \{x, \aleph\}$. At the start of any period t, the agent must choose whether to experience x immediately (i.e. at date t), or to delay it ($\mathbf{d}$) and defer the same choice to his future self. For all $t \in \{0, ..., T-1\}$, denote with h^t the unique non-terminal

history of length t, and with z^{t+1} the unique terminal history of length $t+1$. For all $t \in \{0, ..., T-2\}$, we have $\mathcal{A}_i(h^t) = \{t+1, \mathbf{d}\}$. The agent cannot delay the experience beyond the last feasible date, so $\mathcal{A}_i(h^{T-1}) = \{T\}$.[31] Throughout all the waiting periods, the agent receives nothing. See Lemma 8.2-8.3 for a derivation of the decision and continuation utilities induced by $(\star)$ in $G(T)$.

In what follows, we compare predictions in the game with commitment and discretion. Because we study two games at the same time, we clarify our notation. We use the index K to denote every set or function that refers to $G_K(T)$. Every set or function without the index K refers to $G(T)$ by default. So, for instance, $\mathbf{\Delta}_K$ is the set of behavioral beliefs defined on $G_K(T)$, and DP is the set of deterministic plans defined on $G(T)$. Since chance is inactive, $\Delta^c = \{\sigma_w^c\}$, where each component of σ_w^c is a Dirac measure assigning probability 1 to the waiting action.

For all $t \in \{0, ..., T-2\}$, let $\sigma_1^i(t+1|h^t) := 1$ and $\sigma_T^i(\mathbf{d}|h^t) := 1$. Clearly, σ_1^i (resp. σ_T^i) is a deterministic plan prescribing immediate experience (resp. maximum delay).

Proposition 6.1. *Fix some arbitrary $T > 2$, $\theta \in \text{int}(\Theta)$, $\boldsymbol{\sigma} \in \mathbf{\Delta}_K$, and $q < 0$. There exist thresholds $\underline{\theta}_{\mathcal{D}} \in \mathbb{R}_{++}$ and $\overline{\theta}_{\mathcal{D}} \in [\underline{\theta}_{\mathcal{D}}, \infty)$ such that:*

- *If the agent has a low sensitivity to dread, he prefers maximum delay in $G_K(T)$. Moreover, he is self consistent in $G(T)$, which admits σ_T^i as the unique rational plan.*

$$\theta_{\mathcal{D}} < \underline{\theta}_{\mathcal{D}} \implies \begin{bmatrix} r_{\varnothing}^K(\theta, \boldsymbol{\sigma}) = \{T\} \\ \text{RP}(\theta, \sigma_w^c) = \text{UP}(\theta, \sigma_w^c) \\ \text{RP}(\theta, \sigma_w^c) = \{\sigma_T^i\} \end{bmatrix}$$

- *If the agent has a high sensitivity to dread, he prefers immediate experience in $G_K(T)$. Moreover, he is self consistent in $G(T)$, which admits σ_1^i as the unique rational plan.*

$$\theta_{\mathcal{D}} > \overline{\theta}_{\mathcal{D}} \implies \begin{bmatrix} r_{\varnothing}^K(\theta, \boldsymbol{\sigma}) = \{1\} \\ \text{RP}(\theta, \sigma_w^c) = \text{UP}(\theta, \sigma_w^c) \\ \text{RP}(\theta, \sigma_w^c) = \{\sigma_1^i\} \end{bmatrix}$$

- *If the agent has an intermediate sensitivity to dread, he prefers maximum delay in $G_K(T)$. Moreover, he is dynamically inconsistent in $G(T)$, which admits σ_1^i as the unique rational plan.*

$$\underline{\theta}_{\mathcal{D}} < \theta_{\mathcal{D}} < \overline{\theta}_{\mathcal{D}} \implies \begin{bmatrix} r_{\varnothing}^K(\theta, \boldsymbol{\sigma}) = \{T\} \\ \text{UP}(\theta, \sigma_w^c) = \emptyset \\ \text{RP}(\theta, \sigma_w^c) = \{\sigma_1^i\} \end{bmatrix}$$

If the agent has a low sensitivity to dread, his emotions are not strong enough to subvert the standard predictions of discounted utility theory. All of his contingent selves agree to delay x as much as possible, since they can tolerate the dread experienced during the waiting periods. Conversely, if the agent has a high sensitivity to dread, he wishes to avoid it so strongly that he prefers not to delay x even by a

[31] This means that the agent is inactive at history h^{T-1}. We denote the only available action with T instead of w for notational convenience.

single period.

The most interesting case arises when the agent has an intermediate sensitivity to dread. Here, his tolerance for dread is high enough that, could he commit to his initial choice, he would choose maximum delay. However, it is low enough that he prefers immediate experience over a single period of delay. Suppose that the agent plans to wait until the end. As time passes, dread intensifies, peaking just before the planned date of x. The contingent self who experiences the greatest dread is the last one required to wait (i.e. the h^{T-2}-self, who exists in period $T-1$, acts at h^{T-2} and receives y_{T-1}). Even with an intermediate sensitivity, his dread is so intense that he prefers to experience x immediately (at date $T-1$).

Under commitment he would not be allowed to do so, and would have to wait one final period. Under discretion, however, he is free to experience x immediately, defying the plan of waiting until the end. Consequently, a plan that prescribes maximum delay cannot be rational. The same reasoning applies to any other plan that delays x beyond the first date: the last contingent self required to wait has the same incentive to defy the plan and avoid the greatest dread.

The act of planning unravels: the initial self cannot plan to experience x at date T, because his h^{T-2}-self would experience it at date $T-1$; he cannot plan to experience it at date $T-1$, because his h^{T-3}-self would experience it at date $T-2$; and so on. The initial self understands that none of his future selves would let the game advance past themselves, precisely because they know that every subsequent self would do the same. As a result, the only possible rational plan prescribes immediate experience.

Finally, one can verify (see proof of Proposition 6.1) that:

$$\underline{\theta}_{\mathcal{D}} := \frac{1-\delta}{\phi} \qquad \overline{\theta}_{\mathcal{D}} := \frac{1-\delta^{T-1}}{\sum_{j=1}^{T-1} \delta^{T-j-1}\phi^j}$$

Simple algebra shows that $\underline{\theta}_{\mathcal{D}} < \overline{\theta}_{\mathcal{D}}$, that $\underline{\theta}_{\mathcal{D}}$ and $\overline{\theta}_{\mathcal{D}}$ are both decreasing in ϕ and δ, and that $\overline{\theta}_{\mathcal{D}}$ is increasing in T. This allows us to cleanly perform comparative statics on the horizon, vividness coefficient and discount factor.

For all $t \in \{0, ..., T-2\}$ and $n \in \mathbb{N}$, define:

$$\sigma^i_{\text{mod}(n)}(t+1|h^t) := \begin{cases} 1 & \text{if} \quad (t+1) \bmod n = T \bmod n \\ 0 & \text{if} \quad (t+1) \bmod n \neq T \bmod n \end{cases}$$

Clearly, $\sigma^i_{\text{mod}(n)}$ is a deterministic plan. This plan divides the game into cycles of equal length (except the first cycle, which might be shorter), and prescribes to experience the outcome at the end of each cycle, conditional on not having yet reached a terminal history. If $\sigma^i_{\text{mod}(n)}$ is enacted, the outcome is experienced at the end of the first cycle, i.e. date $\min\{\tau \in \mathbb{T} \colon \tau \bmod n = T \bmod n\}$.

Proposition 6.2. *Fix some arbitrary $T > 2$, $\theta \in \text{int}(\Theta)$, $\boldsymbol{\sigma} \in \boldsymbol{\Delta}_K$, and $q > 0$. There exist thresholds $\underline{\theta}_{\mathcal{S}} \in \mathbb{R}_{++}$ and $\overline{\theta}_{\mathcal{S}} \in [\underline{\theta}_{\mathcal{S}}, \infty]$ such that:*

- *If the agent has a low sensitivity to savoring, he prefers immediate experience in $G_K(T)$. Moreover,*

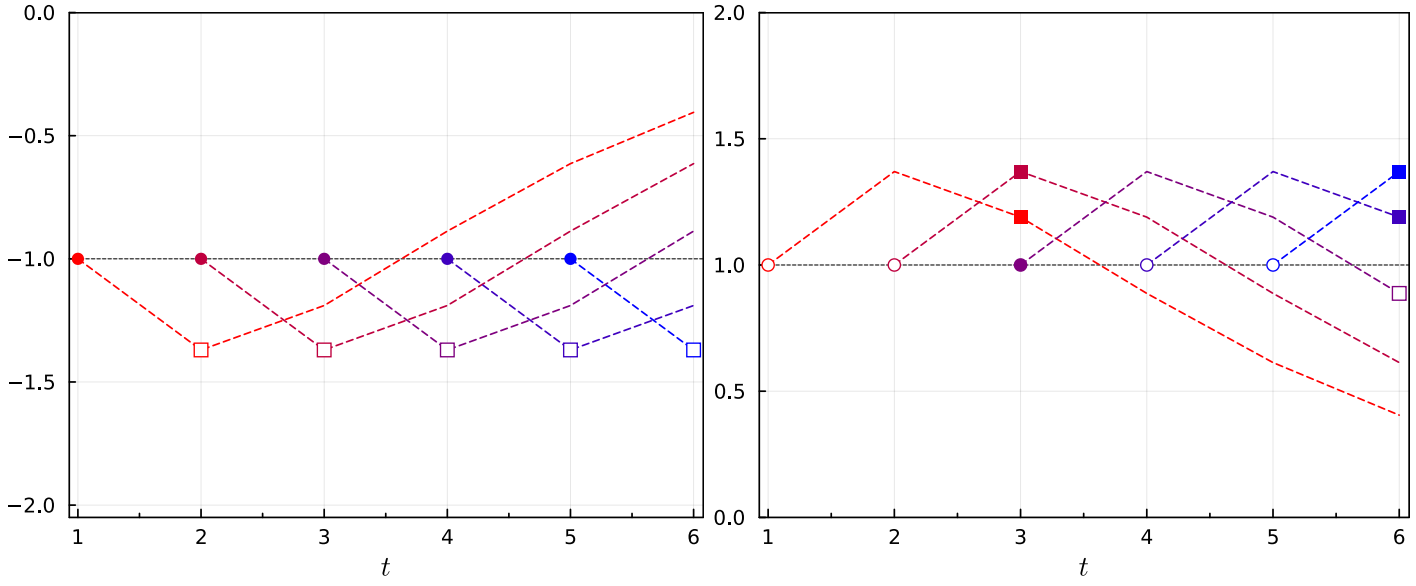


Figure 7: Visualization of $\widehat{U}(t, h^{t-1}, \theta, \sigma_*^i)$ (dot) and $\widehat{U}(\mathbf{d}, h^{t-1}, \theta, \sigma_*^i)$ (square) for $\sigma_*^i \in \mathrm{RP}(\theta, \sigma_w^c)$, $T = 6$, and t increasing from 1 (red) to 5 (blue). On the left, $q = -1$. On the right, $q = 1$. The filled-in symbols mark the best reply of each contingent self. In the case of dread, the outcome would be experienced at date 6 under commitment, and date 1 under discretion. In the case of savoring, the outcome would be experienced at date 2 under commitment, and date 3 under discretion. Here, the contrarian is the h^2-contingent self, so $n^* = T - 1 - 2 = 3$.

he is self consistent in $G(T)$, which admits σ_1^i as the unique rational plan.

$$\theta_{\mathcal{S}} < \underline{\theta}_{\mathcal{S}} \implies \begin{bmatrix} r_{\varnothing}^K(\theta, \boldsymbol{\sigma}) = \{1\} \\ \mathrm{RP}(\theta, \sigma_w^c) = \mathrm{UP}(\theta, \sigma_w^c) \\ \mathrm{RP}(\theta, \sigma_w^c) = \{\sigma_1^i\} \end{bmatrix}$$

- *If the agent has a high sensitivity to savoring, he prefers maximum delay in $G_K(T)$. Moreover, he is self consistent in $G(T)$, which admits σ_T^i as the unique rational plan.*

$$\theta_{\mathcal{S}} > \overline{\theta}_{\mathcal{S}} \implies \begin{bmatrix} r_{\varnothing}^K(\theta, \boldsymbol{\sigma}) = \{T\} \\ \mathrm{RP}(\theta, \sigma_w^c) = \mathrm{UP}(\theta, \sigma_w^c) \\ \mathrm{RP}(\theta, \sigma_w^c) = \{\sigma_T^i\} \end{bmatrix}$$

- *If the agent has an intermediate sensitivity to savoring, he prefers some intermediate delay in $G_K(T)$. Moreover, he is dynamically inconsistent in $G(T)$, which (except for some value $\theta_{\mathcal{S}}^* \in \mathbb{R}_{++}$) admits $\sigma_{\mathrm{mod}(n^*)}^i$ as the unique rational plan, for some $n^* \in \mathbb{N}$.*

$$\underline{\theta}_{\mathcal{S}} < \theta_{\mathcal{S}} < \overline{\theta}_{\mathcal{S}} \implies \begin{bmatrix} 1, T \notin r_{\varnothing}^K(\theta, \boldsymbol{\sigma}) \\ \mathrm{UP}(\theta, \sigma_w^c) = \emptyset \\ \theta_{\mathcal{S}} \neq \theta_{\mathcal{S}}^* \implies \exists n^* \in \mathbb{N} \colon \mathrm{RP}(\theta, \sigma_w^c) = \{\sigma_{\mathrm{mod}(n^*)}^i\} \end{bmatrix}$$

If the agent has a low sensitivity to savoring, his emotions are not strong enough to subvert the standard predictions of discounted utility theory. The agent derives so little enjoyment from savoring that he prefers not to delay x even by a single period. Conversely, if the agent has a high sensitivity to

savoring, he enjoys it so much that all of his contingent selves agree to delay x as much as possible.

The most interesting case arises when the agent has an intermediate sensitivity to savoring. Here, his enjoyment of savoring is high enough that he prefers to delay x. However, it is low enough that, could he commit to his initial choice, he would choose not to delay it all the way. Suppose that the agent plans to wait until his (ex-ante) favorite date. As time passes, savoring intensifies, peaking just before the planned date of x. The contingent self who experiences the greatest savoring is the last one required to wait. Even with an intermediate sensitivity, his savoring is so intense that he would prefer to delay x by one more period.

In the most extreme case, the act of planning unravels: each contingent self keeps delaying, regardless of the plan, so the only rational option is maximum delay. When this is the case, $n^* \geq T$, so $\sigma^i_{\text{mod}(n^*)} = \sigma^i_T$. More often, the logic is subtler. One contingent self (call him a contrarian) may prefer to experience x immediately, knowing that every subsequent self would delay until the final date; a previous contingent self may also prefer to experience x immediately, knowing that every subsequent self would delay until the next contrarian; and so on. Contrarians are evenly spaced, which motivates the use of mod in $\sigma^i_{\text{mod}(n^*)}$. The exact value of n^* is computed by taking $T-1$ and subtracting the length of the history where the last contrarian is found. Further details are explained in the proof.

This dynamic does not unravel completely, but still constrains the amount of delay that can be rationally implemented under discretion. The experience can be delayed up to a specific date, which is far enough into the future to be better than immediate experience, but not so far as to unravel into maximum delay. This date is generally different from the one chosen under commitment, although they may accidentally coincide. The subset of Θ for which this happens is not negligible (not even asymptotically).[32]

Remark 6.3. *In the case of dread ($q < 0$, $T > 2$, $\theta_{\mathcal{D}} \notin \{\underline{\theta}_{\mathcal{D}}, \overline{\theta}_{\mathcal{D}}\}$), a rational agent is dynamically consistent in $G(T)$ iff he chooses the same amount of delay in $G(T)$ and $G_K(T)$.*

Remark 6.3 follows directly from Proposition 6.1. It provides a behavioral characterization of dynamic consistency. The characterization is partial in the sense that it applies only to the case of dread, not savoring: in games with savoring, rational agents might appear dynamically consistent (i.e. choose the same delay under commitment and discretion) even though they are not.

To summarize, we have applied Theorem 2 to extend Loewenstein's (1987) analysis of intertemporal choice with anticipatory emotions from commitment to discretion. Because the game with commitment satisfies OPI, Loewenstein's approach did not require our results on PCPI; however, these results are essential for solving the problem under discretion, which does not satisfy OPI. Thus, although the machinery developed in the first half of the paper is not needed to prove Propositions 5.1 and 5.2, it is central to the proofs of Propositions 6.1 and 6.2.

[32] Even as $T \to \infty$, its measure does not converge to 0, due to the discrete nature of time in our framework.

7. Acknowledgments

I am deeply grateful to Pierpaolo Battigalli for invaluable mentorship and guidance. I also thank Arjada Bardhi, Alberto Bisin, Angelica Capri, Joyee Deb, Guillaume Fréchette, Marco Ottaviani, David Pearce, Debraj Ray, Ariel Rubinstein, Sevgi Yuksel, and two anonymous referees for helpful comments.

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Appendix A. Proofs of Section 4.

Proof of Theorem 2.

We prove the theorem by constructing a deterministic plan $\sigma^i_* \in \Delta^i$, and showing that it is rational. To construct σ^i_*, we use a folding-back algorithm (like a sort of intrapersonal backward induction). Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. For all $k \in \{1, ..., LT\}$, let $H_{LT-k} := \{h \in H \colon \lambda(h) = LT - k\}$.

Initial step.

- Select an arbitrary $\sigma^i_0 \in \Delta^i(0) := \Delta^i$.
- For each $h \in H_{LT-1}$, select an arbitrary $a^*_h \in r_h(\theta, \sigma^i_0, \sigma^c)$.
- For each $h \in H_{LT-1}$, let $\sigma^i_*(a^*_h|h) := 1$.
- Let $\Delta^i(1) := \{\sigma^i \in \Delta^i \colon \forall h \in H_{LT-1},\ \sigma^i(a^*_h|h) = 1\} \neq \emptyset$.

Iterative steps, $k \in \{2, ..., LT\}$. Skip these steps if $LT = 1$.

- Select an arbitrary $\sigma^i_{k-1} \in \Delta^i(k-1)$.
- For each $h \in H_{LT-k}$, select an arbitrary $a^*_h \in r_h(\theta, \sigma^i_{k-1}, \sigma^c)$.
- For each $h \in H_{LT-k}$, let $\sigma^i_*(a^*_h|h) := 1$.

- Let $\Delta^i(k) := \{\sigma^i \in \Delta^i \colon \forall h \in \bigcup_{n=1}^{k} H_{LT-n},\ \sigma^i(a_h^*|h) = 1\} \neq \emptyset$.

By construction, $\sigma_*^i \in \text{DP}$ and $\sigma_*^i \in \Delta^i(LT)$. For all $k \in \{1, ..., LT\}$, $h \in H_{LT-k}$, and $\sigma^i \in \Delta^i(k-1)$, by PCPI we have $r_h(\theta, \sigma^i, \sigma^c) = r_h(\theta, \sigma_{k-1}^i, \sigma^c)$. Notice that $(\Delta^i(k))_{k=0}^{LT}$ is a decreasing sequence of sets, so $\sigma_*^i \in \Delta^i(k-1)$ for all $k \in \{1, ..., LT\}$. But then, for all $h \in \bigcup_{n=1}^{LT} H_{LT-n} \equiv H$ we have supp $\sigma_*^i(\cdot|h) = \{a_h^*\} \subseteq r_h(\theta, \sigma_*^i, \sigma^c)$. Therefore, $\sigma_*^i \in \text{RP}(\theta, \sigma^c)$. □

Lemma 7.1. *Under PCPI, for all $h \in H$, $\theta \in \Theta$, and $\sigma^c \in \Delta^c$, the following holds:*

$$\max_{a_i \in \mathcal{A}_i(h)} \left[\max_{\sigma^i \in \Delta^i} \widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c) \right] = \max_{\sigma^i \in \Delta^i} \vec{U}(h, \theta, \sigma^i, \sigma^c)$$

Proof. Fix some arbitrary $h \in H$, $\theta \in \Theta$, and $\sigma^c \in \Delta^c$. Under PCPI, the following holds:

$$\begin{aligned}
\max_{a_i \in \mathcal{A}_i(h)} \left[\max_{\sigma^i \in \Delta^i} \widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c) \right] &= \max_{\sigma^i \in \Delta^i} \left[\max_{a_i \in \mathcal{A}_i(h)} \widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c) \right] \\
&= \max_{\sigma^i \in \Delta^i} \left[\max_{\widehat{\sigma}^i \in \Delta^i} \sum_{a_i \in \mathcal{A}_i(h)} \widehat{\sigma}^i(a_i|h) \cdot \widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c) \right] \\
&= \max_{\sigma^i \in \Delta^i} \sum_{a_i \in \mathcal{A}_i(h)} \sigma^i(a_i|h) \cdot \widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c) \\
&= \max_{\sigma^i \in \Delta^i} \vec{U}(h, \theta, \sigma^i, \sigma^c)
\end{aligned}$$

The third equality holds by PCPI. Indeed, $\widehat{U}(a_i, h, \theta, \sigma^i, \sigma^c)$ does not depend on $\sigma^i(\cdot|h)$, so the maximization over $\widehat{\sigma}^i(\cdot|h)$ can be absorbed into the maximization over σ^i. □

Proof of Lemma 2.1.

Fix some arbitrary $\theta \in \Theta$ and $\sigma^c \in \Delta^c$. If $\text{UP}(\theta, \sigma^c) = \emptyset$, then clearly $\text{UP}(\theta, \sigma^c) \subseteq \text{RP}(\theta, \sigma^c)$. Suppose instead that $\text{UP}(\theta, \sigma^c) \neq \emptyset$. Fix an arbitrary $\sigma_*^i \in \text{UP}(\theta, \sigma^c)$. We want to show that $\sigma_*^i \in \text{RP}(\theta, \sigma^c)$. The following holds:

$$\vec{U}(h, \theta, \sigma_*^i, \sigma^c) = \max_{\sigma^i \in \Delta^i} \vec{U}(h, \theta, \sigma^i, \sigma^c) \geq \max_{a_i \in \mathcal{A}_i(h)} \widehat{U}(a_i, h, \theta, \sigma_*^i, \sigma^c) \geq \vec{U}(h, \theta, \sigma_*^i, \sigma^c)$$

The equality holds by definition of sequential optimality, and the first inequality holds by PCPI and Lemma 7.1. Therefore, we obtain:

$$\max_{a_i \in \mathcal{A}_i(h)} \widehat{U}(a_i, h, \theta, \sigma_*^i, \sigma^c) = \vec{U}(h, \theta, \sigma_*^i, \sigma^c) = \sum_{a_i \in \mathcal{A}_i(h)} \sigma_*^i(a_i|h) \cdot \widehat{U}(a_i, h, \theta, \sigma_*^i, \sigma^c)$$

This implies that σ_*^i is one-step optimal, meaning that $\sigma_*^i \in \text{RP}(\theta, \sigma^c)$. □

Proof of Corollary 2.2.

Fix some arbitrary $\theta \in \Theta$, $\sigma^c \in \Delta^c$, and $\sigma_*^i \in \text{RP}(\theta, \sigma^c)$. Let $H^* := \{h \in H \colon |r_h(\theta, \sigma_*^i, \sigma^c)| > 1\}$.

"If" side. We proceed by contraposition. Suppose that $H^* \neq \emptyset$. We want to show that there exists some $\sigma_{**}^i \in \text{RP}(\theta, \sigma^c)$ such that $\sigma_{**}^i \neq \sigma_*^i$. Let $k^* := LT - \max_{h \in H^*} \lambda(h)$. We follow a variation of the algorithm used in the proof of Theorem 2.

Initial step. Skip this step if $k^* = 1$.

- For all $h \in \bigcup_{n=1}^{k^*-1} H_{LT-n}$, let $\sigma^i_{**}(\cdot|h) := \sigma^i_*(\cdot|h)$.
- Let $\Delta^i_*(k^*-1) := \{\sigma^i \in \Delta^i \colon \forall h \in \bigcup_{k=1}^{k^*-1} H_{LT-k},\ \sigma^i(\cdot|h) = \sigma^i_{**}(\cdot|h)\}$.

Crucial step. Start here if $k^* = 1$, and let $\Delta^i_*(0) = \Delta^i$.

- Select an arbitrary $\sigma^i_{k^*-1} \in \Delta^i_*(k^*-1)$.
- Select an arbitrary $h^* \in H^* \cap H_{LT-k^*} \neq \emptyset$.
- For each $h \in H_{LT-k^*} \setminus \{h^*\}$, let $\sigma^i_{**}(\cdot|h) \in \Delta(r_h(\theta, \sigma^i_{k^*-1}, \sigma^c))$.
- Let $\sigma^i_{**}(\cdot|h^*) \in \Delta(r_{h^*}(\theta, \sigma^i_{k^*-1}, \sigma^c))$, with $\sigma^i_{**}(\cdot|h^*) \neq \sigma^i_*(\cdot|h^*)$. This is possible since $|r_{h^*}(\theta, \sigma^i_{k^*-1}, \sigma^c)| = |r_{h^*}(\theta, \sigma^i_*, \sigma^c)| > 1$.
- Let $\Delta^i_*(k^*) := \{\sigma^i \in \Delta^i \colon \forall h \in \bigcup_{n=1}^{k^*} H_{LT-n},\ \sigma^i(\cdot|h) = \sigma^i_{**}(\cdot|h)\} \neq \emptyset$.

Iterative steps, $k \in \{k^*+1, ..., LT\}$. Skip these steps if $k^* = LT$.

- Select an arbitrary $\sigma^i_{k-1} \in \Delta^i_*(k-1)$.
- For each $h \in H_{LT-k}$, let $\sigma^i_{**}(\cdot|h) \in \Delta(r_h(\theta, \sigma^i_{k-1}, \sigma^c))$.
- Let $\Delta^i_*(k) := \{\sigma^i \in \Delta^i \colon \forall h \in \bigcup_{n=1}^{k} H_{LT-n},\ \sigma^i(\cdot|h) = \sigma^i_{**}(\cdot|h)\} \neq \emptyset$.

By construction, $\sigma^i_{**} \neq \sigma^i_*$ and $\sigma^i_{**} \in \Delta^i_*(LT)$. For all $k \in \{k^*, ..., LT\}$, $h \in H_{LT-k}$, and $\sigma^i \in \Delta^i_*(k-1)$, by PCPI we have $r_h(\theta, \sigma^i, \sigma^c) = r_h(\theta, \sigma^i_{k-1}, \sigma^c)$. Notice that $(\Delta^i_*(k))_{k=k^*-1}^{LT}$ is a decreasing sequence of sets, so $\sigma^i_{**} \in \Delta^i_*(k-1)$ for all $k \in \{k^*, ..., LT\}$. But then, for all $h \in \bigcup_{n=k^*}^{LT} H_{LT-n}$ we have supp $\sigma^i_{**}(\cdot|h) \subseteq r_h(\theta, \sigma^i_{**}, \sigma^c)$. Moreover (if $k^* > 1$), for all $h \in \bigcup_{n=1}^{k^*-1} H_{LT-n}$ we have supp $\sigma^i_{**}(\cdot|h) =$ supp $\sigma^i_*(\cdot|h) \subseteq r_h(\theta, \sigma^i_*, \sigma^c) = r_h(\theta, \sigma^i_{**}, \sigma^c)$. Therefore, $\sigma^i_{**} \in \mathrm{RP}(\theta, \sigma^c)$.

"Only if" side. We proceed by contraposition. Suppose that there exists some $\sigma^i_{**} \in \mathrm{RP}(\theta, \sigma^c)$ such that $\sigma^i_{**} \neq \sigma^i_*$. We want to show that $H^* \neq \emptyset$. Let $H_* := \{h \in H \colon \sigma^i_{**}(\cdot|h) \neq \sigma^i_*(\cdot|h)\}$. Let $k_* := LT - \max_{h \in H_*} \lambda(h)$. Clearly, $\sigma^i_*(\cdot|h) = \sigma^i_{**}(\cdot|h)$ for all $h \in \bigcup_{n=1}^{k_*-1} H_{LT-n}$. For all $h \in H_{LT-k_*}$, by PCPI we have $r_h(\theta, \sigma^i_{**}, \sigma^c) = r_h(\theta, \sigma^i_*, \sigma^c)$. Select an arbitrary $h_* \in H_* \cap H_{LT-k_*} \neq \emptyset$. Then, $\sigma^i_{**}(\cdot|h_*) \in \Delta(r_{h_*}(\theta, \sigma^i_{**}, \sigma^c)) = \Delta(r_{h_*}(\theta, \sigma^i_*, \sigma^c)) \ni \sigma^i_*(\cdot|h_*)$. Since $\sigma^i_{**}(\cdot|h_*) \neq \sigma^i_*(\cdot|h_*)$, we conclude that $h_* \in H^*$. □

Proof of Proposition 3.4.

Suppose that $|\Delta^c| = 1$. Then $\mathcal{A}_c(h) = \{w\}$ for all $h \in H$, and $\Delta^c = \{\sigma^c_w\}$, where each component of σ^c_w is a Dirac measure assigning probability 1 to the waiting action w. We proceed by contradiction: suppose that the decision problem satisfies PPI but not PCPI. Then, there exists some $h \in H$, $a_i \in \mathcal{A}_i(h)$, $\theta \in \Theta$, and $\widehat{\sigma}^i, \widetilde{\sigma}^i \in \Delta^i$ with $(\widehat{\sigma}^i(\cdot|h'))_{h' \succ h}^{h' \in H} = (\widetilde{\sigma}^i(\cdot|h'))_{h' \succ h}^{h' \in H}$ and $(\widehat{\sigma}^i(\cdot|h'))_{h' \preceq h}^{h' \in H} = (\widetilde{\sigma}^i(\cdot|h'))_{h' \preceq h}^{h' \in H}$ such that $\widehat{U}(a_i, h, \theta, \widehat{\sigma}^i, \sigma^c_w) \neq \widehat{U}(a_i, h, \theta, \widetilde{\sigma}^i, \sigma^c_w)$. Fix an arbitrary $z^* \in Z$ such that $\alpha_i(h, z^*) = a_i$. Let $\widehat{\sigma}^i_*(\cdot|h') := \widehat{\sigma}^i(\cdot|h')$ for all $h' \in H$ such that $h' \not\succeq h$, and let $\widehat{\sigma}^i_*(\alpha_i(h', z^*)|h') := 1$ for all $h' \in H$ such that $h' \preceq h$. By PPI, $\widehat{U}(a_i, h, \theta, \widehat{\sigma}^i, \sigma^c_w) = \widehat{U}(a_i, h, \theta, \widehat{\sigma}^i_*, \sigma^c_w)$. Let $\widetilde{\sigma}^i_*(\cdot|h') := \widetilde{\sigma}^i(\cdot|h')$ for all $h' \in H$ such that $h' \not\succeq h$, and let $\widetilde{\sigma}^i_*(\alpha_i(h', z^*)|h') := 1$ for all $h' \in H$ such that $h' \preceq h$. By PPI, $\widehat{U}(a_i, h, \theta, \widetilde{\sigma}^i, \sigma^c_w) = \widehat{U}(a_i, h, \theta, \widetilde{\sigma}^i_*, \sigma^c_w)$. By construction (and degeneracy of σ^c_w), we have $\mathbb{P}^{h'}_{\widehat{\sigma}^i_*, \sigma^c_w} = \mathbb{P}^{h'}_{\widetilde{\sigma}^i_*, \sigma^c_w}$

for all $h' \in H$ such that $h' \preceq (h,(a_i,w))$ or $h' \succ (h,(a_i,w))$. Consequently, $\psi(z,\widehat{\sigma}^i_*,\sigma^c_w) = \psi(z,\widetilde{\sigma}^i_*,\sigma^c_w)$ for all $z \in Z$ such that $z \succeq (h,(a_i,w))$, so $U_{\mathbf{T}(h,(a_i,w))}(z,\theta,\widehat{\sigma}^i_*,\sigma^c_w) = U_{\mathbf{T}(h,(a_i,w))}(z,\theta,\widetilde{\sigma}^i_*,\sigma^c_w)$ for all $z \in Z$ such that $z \succeq (h,(a_i,w))$. But then $\widehat{U}(a_i,h,\theta,\widehat{\sigma}^i_*,\sigma^c_w) = \widehat{U}(a_i,h,\theta,\widetilde{\sigma}^i_*,\sigma^c_w)$, so we reach the contradiction $\widehat{U}(a_i,h,\theta,\widehat{\sigma}^i,\sigma^c_w) = \widehat{U}(a_i,h,\theta,\widetilde{\sigma}^i,\sigma^c_w)$. □

Appendix B. Proofs of Section 6.

Lemma 8.1. Canonical Interpolation. *Fix some arbitrary $\theta = (\theta_{\mathcal{S}},\theta_{\mathcal{D}},\phi,\delta) \in \text{int}(\Theta)$ and $q \in \mathbb{R}\setminus\{0\}$. Let θ_q be a variable that takes value $\theta_{\mathcal{S}}$ (resp. $\theta_{\mathcal{D}}$) if $q>0$ (resp. $q<0$). Define $f_q\colon \mathbb{N}\to\mathbb{R}_{++}$ as:*

$$f_q(n) := \theta_q \sum_{j=0}^{n-1} \delta^{n-j-1}\phi^j + (1-\theta_q)\delta^{n-1}$$

Define $g_q\colon [1,\infty)\to\mathbb{R}_{++}$ as:

$$g_q(\tau) := \begin{cases} \theta_q \cdot \frac{\delta^\tau - \phi^\tau}{\delta-\phi} + (1-\theta_q)\delta^{\tau-1} & \text{if} \quad \phi \neq \delta \\ \theta_q \cdot \tau\delta^{\tau-1} + (1-\theta_q)\delta^{\tau-1} & \text{if} \quad \phi = \delta \end{cases}$$

Function g_q is:

(1) Strictly quasi-concave:

$$\exists \tau^* \in [1,\infty)\colon \begin{bmatrix} \tau' < \tau'' \leq \tau^* \implies g_q(\tau') < g_q(\tau'') \\ \tau' > \tau'' \geq \tau^* \implies g_q(\tau') < g_q(\tau'') \end{bmatrix}$$

(2) Maximized at a single point: $|\arg\max_{\tau\in[1,\infty)} g_q(\tau)| = 1$.
(3) An interpolation of f_q: $\forall n \in \mathbb{N},\ g_q(n) = f_q(n)$.

Function f_q is:

(4) Asymptotically grounded: $\lim_{n\to\infty} f_q(n) = 0$.
(5) Single-plateaued: $f_q(n) \geq f_q(n+1) \implies f_q(n+1) > f_q(n+2)$.
(6) Maximized at a single point or at two adjacent points:

$$1 \leq |\arg\max_{n\in\mathbb{N}} f_q(n)| \leq 2$$
$$\widehat{n},\widetilde{n} \in \arg\max_{n\in\mathbb{N}} f_q(n),\ \widehat{n} < \widetilde{n} \implies \widehat{n}+1 = \widetilde{n}$$

Proof. (1) If $\phi \neq \delta$, the following holds:

$$\left.\frac{\partial g_q(\tau)}{\partial\tau}\right|_{\tau=\widehat{\tau}} = \theta_q \cdot \frac{\ln(\delta)\delta^{\widehat{\tau}} - \ln(\phi)\phi^{\widehat{\tau}}}{\delta-\phi} + \frac{1-\theta_q}{\delta}\ln(\delta)\delta^{\widehat{\tau}} = 0 \iff \widehat{\tau} = \log_{\phi/\delta}\left(\frac{[\delta-(1-\theta_q)\phi]\ln(\delta)}{\delta\theta_q\ln(\phi)}\right)$$

There is a stationary point if $\widehat{\tau} \in [1,\infty)$, and no stationary points otherwise.
If $\phi = \delta$, the following holds:

$$\left.\frac{\partial g_q(\tau)}{\partial\tau}\right|_{\tau=\widetilde{\tau}} = \theta_q \cdot \delta^{\widetilde{\tau}-1}(1+\widetilde{\tau}\ln(\delta)) + (1-\theta_q)\ln(\delta)\delta^{\widetilde{\tau}-1} = 0 \iff \widetilde{\tau} = 1 - \frac{1}{\theta_q} - \frac{1}{\ln(\delta)}$$

There is a stationary point if $\widetilde{\tau} \in [1, \infty)$, and no stationary points otherwise.

In both cases, $g_q(1) = 1 > 0 = \lim_{\tau \to \infty} g_q(\tau)$. Since g_q is differentiable, it is single-peaked if it has a stationary point, and strictly decreasing otherwise. Either way, it is strictly quasi-concave.

(2) Follows from *(1)* and $g_q(1) = 1 > 0 = \lim_{\tau \to \infty} g_q(\tau)$.

(3) If $\phi = \delta$, then $f_q(n) = \theta_q \sum_{j=0}^{n-1} \delta^{n-1} + (1 - \theta_q)\delta^{n-1} = \theta_q \cdot n\delta^{n-1} + (1 - \theta_q)\delta^{n-1} = g_q(n)$ for all $n \in \mathbb{N}$. If $\phi \neq \delta$, then $\delta[f_q(n) - (1 - \theta_q)\delta^{n-1}] + \theta_q \phi^n = \phi[f_q(n) - (1 - \theta_q)\delta^{n-1}] + \theta_q \delta^n$ for all $n \in \mathbb{N}$, so $f_q(n) = \theta_q \cdot \frac{\delta^n - \phi^n}{\delta - \phi} + (1 - \theta_q)\delta^{n-1} = g_q(n)$.

(4) Follows from *(3)* and $\lim_{\tau \to \infty} g_q(\tau) = 0$.

(5) Follows from *(1)*, *(2)*, and *(3)*.

(6) Follows from *(4)*, *(5)* and the contrapositive of *(5)*. □

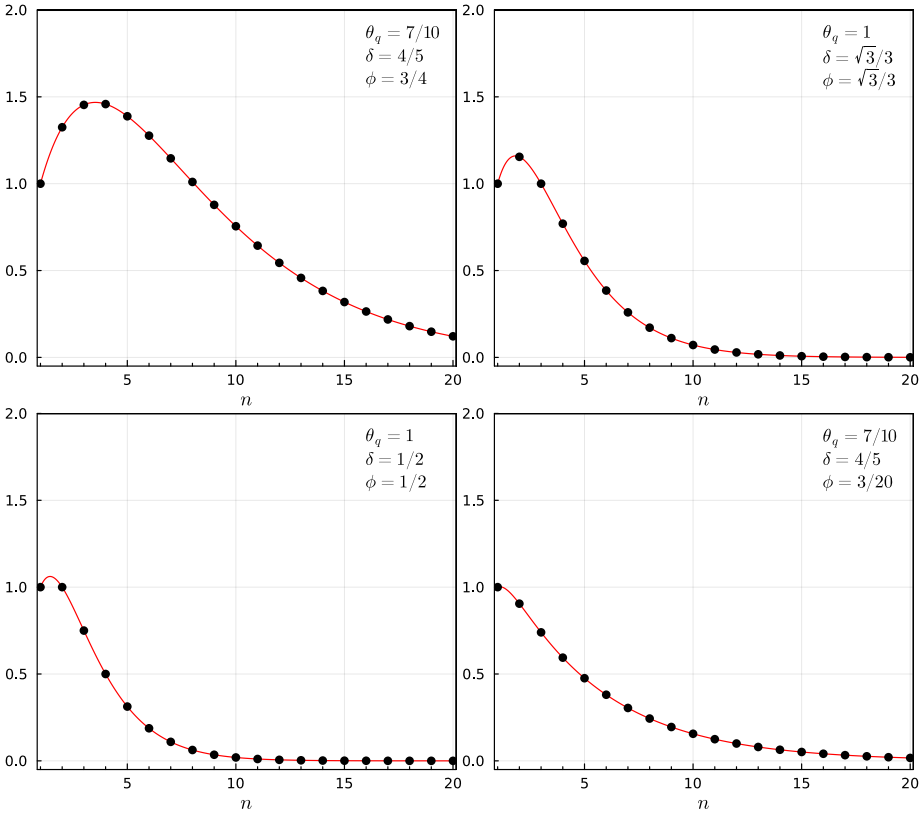


Figure B.8: Visualization of f_q (black) and g_q (red) for different values of θ_q, ϕ, δ.

Proof of Proposition 5.1.

Fix some arbitrary $T \in \mathbb{N}$, $\theta \in \text{int}(\Theta)$, $\boldsymbol{\sigma} \in \boldsymbol{\Delta}$, and $q < 0$. Let θ_q and f_q be defined as in Lemma 8.1. If $T = 1$, the result is trivial. Suppose instead that $T > 1$. We can write:

$$\widehat{U}(a_i, \varnothing, \theta, \boldsymbol{\sigma}) = \begin{cases} q & \text{if} \quad a_i = 1 \\ [\theta_{\mathcal{D}} \cdot \phi + \delta] \cdot q & \text{if} \quad a_i = 2 \\ \vdots & \vdots \\ [\theta_{\mathcal{D}} \cdot (\phi^{T-1} + \delta\phi^{T-2} + \delta^2\phi^{T-3} + ... + \delta^{T-2}\phi) + \delta^{T-1}] \cdot q & \text{if} \quad a_i = T \end{cases}$$
$$= \left[\theta_{\mathcal{D}} \sum_{j=0}^{a_i-1} \delta^{a_i-j-1}\phi^j + (1-\theta_{\mathcal{D}})\delta^{a_i-1}\right] \cdot q = f_q(a_i) \cdot q$$

Since $q < 0$, $\widehat{U}$ is maximized over $\mathcal{A}_i(\varnothing)$ as f_q is minimized over $\mathbb{T} = \mathcal{A}_i(\varnothing)$. Formally, $r_{\varnothing}(\theta, \boldsymbol{\sigma}) = \arg\min_{n \in \mathbb{T}} f_q(n)$. By *(5)* of Lemma 8.1, if $T \leq \widetilde{n} := \max\{\arg\max_{n \in \mathbb{N}} f_q(n)\}$, then $\arg\min_{n \in \mathbb{T}} f_q(n) = \{1\}$. If $T > \widetilde{n}$, then $\arg\min_{n \in \{1,...,\widetilde{n}\}} f_q(n) = \{1\}$ and $\arg\min_{n \in \{\widetilde{n}+1,...,T\}} f_q(n) = \{T\}$. Therefore, $\arg\min_{n \in \mathbb{T}} f_q(n) \subseteq \{1, T\}$. □

Proof of Proposition 5.2.

Fix some arbitrary $T \in \mathbb{N}$, $\theta \in \text{int}(\Theta)$, $\boldsymbol{\sigma} \in \boldsymbol{\Delta}_K$, and $q > 0$. Let θ_q and f_q and g_q be defined as in Lemma 8.1. If $T = 1$, the result is trivial. Suppose instead that $T > 1$. We can write:

$$\widehat{U}(a_i, \varnothing, \theta, \boldsymbol{\sigma}) = \begin{cases} q & \text{if} \quad a_i = 1 \\ [\theta_{\mathcal{S}} \cdot \phi + \delta] \cdot q & \text{if} \quad a_i = 2 \\ \vdots & \vdots \\ [\theta_{\mathcal{S}} \cdot (\phi^{T-1} + \delta\phi^{T-2} + \delta^2\phi^{T-3} + ... + \delta^{T-2}\phi) + \delta^{T-1}] \cdot q & \text{if} \quad a_i = T \end{cases}$$
$$= \left[\theta_{\mathcal{S}} \sum_{j=0}^{a_i-1} \delta^{a_i-j-1}\phi^j + (1-\theta_{\mathcal{S}})\delta^{a_i-1}\right] \cdot q = f_q(a_i) \cdot q$$

Since $q > 0$, $\widehat{U}$ is maximized over $\mathcal{A}_i(\varnothing)$ as f_q is maximized over $\mathbb{T} = \mathcal{A}_i(\varnothing)$. Formally, $r_{\varnothing}(\theta, \boldsymbol{\sigma}) = \arg\max_{n \in \mathbb{T}} f_q(n)$. By *(3)* of Lemma 8.1, if $\arg\max_{\tau \in [1,\infty)} g_q(\tau) = \{t\} \subset \mathbb{N}$, then $r_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{t\}$. Recall from the proof of Lemma 8.1 *(1)* that for all $t \in \mathbb{N}$, if $\delta = \phi$ and $t = 1 - \frac{1}{\theta_q} - \frac{1}{\ln(\delta)}$ then $\arg\max_{\tau \in [1,\infty)} g_q(\tau) = \{t\}$. Thus, for all $t \in \mathbb{N}$, if $\theta = (1, 1, e^{-1/t}, e^{-1/t})$ then $r_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{t\}$. □

Lemma 8.2. *Fix an arbitrary $T > 2$ and consider $G(T)$, from Section 6.2. Fix some arbitrary $q \in \mathbb{R} \setminus \{0\}$, $\theta \in \text{int}(\Theta)$ and $\sigma^i \in \text{DP}$. Let θ_q and f_q be defined as in Lemma 8.1. For all $t \in \{0, ..., T-2\}$, there exist numbers $p^1_{\sigma^i}(t), p^2_{\sigma^i}(t) \in \mathbb{T}$ such that:*

$$\widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i, \sigma^c_w) = \begin{cases} f_q(p^1_{\sigma^i}(t) - t) \cdot q & \text{if} \quad p^1_{\sigma^i}(t) > t+1 \\ \\ f_q(p^2_{\sigma^i}(t) - t) \cdot q & \text{if} \quad p^1_{\sigma^i}(t) = t+1 \end{cases}$$

Proof. For all $t \in \{0, ..., T-1\}$ and $\sigma^i \in \text{DP}$, let $p^1_{\sigma^i}(t)$ be the date at which the agent experiences x if he follows σ^i starting from h^t. Formally:

$$p^1_{\sigma^i}(t) := \min\{\tau \in \{t+1, ..., T\} \colon \sigma^i(\tau | h^{\tau-1}) = 1\}$$

For all $t \in \{0, ..., T-1\}$ and $\sigma^i \in \text{DP}$, let $p^2_{\sigma^i}(t)$ be the date at which the agent experiences x if he follows σ^i starting from $h^{p^1_{\sigma^i}(t)}$. Formally:

$$p^2_{\sigma^i}(t) := \begin{cases} \min\{\tau \in \{p^1_{\sigma^i}(t)+1, ..., T\} \colon \sigma^i(\tau | h^{\tau-1}) = 1\} & \text{if} \quad p^1_{\sigma^i}(t) < T \\ T & \text{if} \quad p^1_{\sigma^i}(t) = T \end{cases}$$

For all $t \in \{0, ..., T-1\}$, we have $p^1_{\sigma^i}(t) \leq p^2_{\sigma^i}(t)$. Moreover, $p^1_{\sigma^i}(t) = p^2_{\sigma^i}(t)$ only if $p^1_{\sigma^i}(t) = T$. Finally, for all $t \in \{0, ..., T-2\}$, the following holds:

$$\begin{aligned} p^1_{\sigma^i}(t) > t+1 &\implies p^1_{\sigma^i}(t+1) = p^1_{\sigma^i}(t) \\ p^1_{\sigma^i}(t) > t+1 &\implies p^2_{\sigma^i}(t+1) = p^2_{\sigma^i}(t) \\ p^1_{\sigma^i}(t) = t+1 &\implies p^1_{\sigma^i}(t+1) = p^2_{\sigma^i}(t) \end{aligned}$$

Because σ^i is a deterministic plan, knowing $p^1_{\sigma^i}(t)$ and $p^2_{\sigma^i}(t)$ is sufficient to compute the value of $\widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i, \sigma^c_w)$. By extrapolation from Table B.3, convince yourself of the following. For all $t \in \{0, ..., T-2\}$, we have:

$$\widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i, \sigma^c_w) = \begin{cases} \left[\theta_q \sum_{x=0}^{p^1_{\sigma^i}(t)-t-1} \delta^{p^1_{\sigma^i}(t)-t-x-1}\phi^x + (1-\theta_q)\delta^{p^1_{\sigma^i}(t)-t-1}\right] \cdot q & \text{if} \quad p^1_{\sigma^i}(t) > t+1 \\ \left[\theta_q \sum_{x=0}^{p^2_{\sigma^i}(t)-t-1} \delta^{p^2_{\sigma^i}(t)-t-x-1}\phi^x + (1-\theta_q)\delta^{p^2_{\sigma^i}(t)-t-1}\right] \cdot q & \text{if} \quad p^1_{\sigma^i}(t) = t+1 \end{cases}$$

$$= \begin{cases} f_q(p^1_{\sigma^i}(t) - t) \cdot q & \text{if} \quad p^1_{\sigma^i}(t) > t+1 \\ f_q(p^2_{\sigma^i}(t) - t) \cdot q & \text{if} \quad p^1_{\sigma^i}(t) = t+1 \end{cases}$$

□

Lemma 8.3. *Fix an arbitrary $T > 2$ and consider $G(T)$, from Section 6.2. Fix some arbitrary $q \in \mathbb{R} \setminus \{0\}$, $\theta \in \text{int}(\Theta)$ and $\sigma^i \in \Delta^i$. Let θ_q and f_q be defined as in Lemma 8.1. For all $t \in \{0, ..., T-2\}$, we have:*

$$\vec{U}(z^{t+1}, \theta, \sigma^i, \sigma^c_w) = \vec{U}(z^T, \theta, \sigma^i, \sigma^c_w) = \vec{U}(h^{T-1}, \theta, \sigma^i, \sigma^c_w) = q$$

$$\vec{U}(h^t, \theta, \sigma^i, \sigma^c_w) = \sum_{\tau=t+1}^{T} \mathbb{P}^{h^t}_{\sigma^i, \sigma^c_w}(z^\tau) \cdot f_q(\tau - t) \cdot q$$

Proof. Fix an arbitrary $t \in \{0, .., T-2\}$. The first line is trivial. As for the second line, the following holds:

$$\begin{aligned} \vec{U}(h^t, \theta, \sigma^i, \sigma^c_w) &= \sigma^i(t+1|h^t) \cdot \widehat{U}(t+1, h^t, \theta, \sigma^i, \sigma^c_w) + \sigma^i(\mathbf{d}|h^t) \cdot \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i, \sigma^c_w) \\ &= \sigma^i(t+1|h^t) \cdot q + \sigma^i(\mathbf{d}|h^t) \cdot \sum_{\tau=t+2}^{T} U_{t+1}(z^\tau, \theta, \sigma^i, \sigma^c_w) \cdot \mathbb{P}^{h^{t+1}}_{\sigma^i, \sigma^c_w}(z^\tau) \end{aligned}$$

t	$p^1_{\sigma^i}(t)$	$p^2_{\sigma^i}(t)$	$\widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i, \sigma^c_w)$
0	1	2	$[\theta_q \cdot \phi + \delta] \cdot q$
		3	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		...	...
		T	$[\theta_q \cdot (\phi^{T-1} + \delta\phi^{T-2} + \delta^2\phi^{T-3} + ... + \delta^{T-2}\phi) + \delta^{T-1}] \cdot q$
	2	3	$[\theta_q \cdot \phi + \delta] \cdot q$
		4	$[\theta_q \cdot \phi + \delta] \cdot q$
		...	...
		T	$[\theta_q \cdot \phi + \delta] \cdot q$
	3	4	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		5	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		...	...
		T	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
	...	...	...
	$T-1$	T	$[\theta_q \cdot (\phi^{T-2} + \delta\phi^{T-3} + \delta^2\phi^{T-4} + ... + \delta^{T-3}\phi) + \delta^{T-2}] \cdot q$
	T	T	$[\theta_q \cdot (\phi^{T-1} + \delta\phi^{T-2} + \delta^2\phi^{T-3} + ... + \delta^{T-2}\phi) + \delta^{T-1}] \cdot q$
1	2	3	$[\theta_q \cdot \phi + \delta] \cdot q$
		4	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		...	...
		T	$[\theta_q \cdot (\phi^{T-2} + \delta\phi^{T-3} + \delta^2\phi^{T-4} + ... + \delta^{T-3}\phi) + \delta^{T-2}] \cdot q$
	3	4	$[\theta_q \cdot \phi + \delta] \cdot q$
		5	$[\theta_q \cdot \phi + \delta] \cdot q$
		...	...
		T	$[\theta_q \cdot \phi + \delta] \cdot q$
	4	5	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		6	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		...	...
		T	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
	...	...	...
	$T-1$	T	$[\theta_q \cdot (\phi^{T-3} + \delta\phi^{T-4} + \delta^2\phi^{T-5} + ... + \delta^{T-4}\phi) + \delta^{T-3}] \cdot q$
	T	T	$[\theta_q \cdot (\phi^{T-2} + \delta\phi^{T-3} + \delta^2\phi^{T-4} + ... + \delta^{T-3}\phi) + \delta^{T-2}] \cdot q$
...	...	...	...
t	$t+1$	$t+2$	$[\theta_q \cdot \phi + \delta] \cdot q$
		$t+3$	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		...	...
		T	$[\theta_q \cdot (\phi^{T-t-1} + \delta\phi^{T-t-2} + ... + \delta^{T-t-2}\phi) + \delta^{T-t-1}] \cdot q$
	$t+2$	$t+3$	$[\theta_q \cdot \phi + \delta] \cdot q$
		$t+4$	$[\theta_q \cdot \phi + \delta] \cdot q$
		...	...
		T	$[\theta_q \cdot \phi + \delta] \cdot q$
	$t+3$	$t+5$	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		$t+4$	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
		...	...
		T	$[\theta_q \cdot (\phi^2 + \delta\phi) + \delta^2] \cdot q$
	...	...	...
	$T-1$	T	$[\theta_q \cdot (\phi^{T-t-2} + \delta\phi^{T-t-3} + ... + \delta^{T-t-3}\phi) + \delta^{T-t-2}] \cdot q$
	T	T	$[\theta_q \cdot (\phi^{T-t-1} + \delta\phi^{T-t-2} + ... + \delta^{T-t-2}\phi) + \delta^{T-t-1}] \cdot q$
...	...	...	...
$T-2$	$T-1$	T	$[\theta_q \cdot \phi + \delta] \cdot q$
$T-2$	T	T	$[\theta_q \cdot \phi + \delta] \cdot q$
$T-1$	T	T	Undefined, since $\mathbf{d} \notin \mathcal{A}_i(h^{T-1}) = \{T\}$

Table B.3: Decision utility of playing $\mathbf{d}$ at h^t, while holding the deterministic plan $\sigma^i \in \text{DP}$.

$$
\begin{aligned}
&= \sigma^i(t+1|h^t)\cdot q + \sigma^i(\mathbf{d}|h^t)\cdot\Bigg[\nu_{t+1}(\pi(h^{t+1}),\theta,\psi(h^{t+1},\sigma^i,\sigma^c_w)) \\
&\qquad\qquad + \delta\cdot\sum_{\tau=t+2}^{T} U_{t+2}(z^\tau,\theta,\sigma^i,\sigma^c_w)\cdot\mathbb{P}^{h^{t+1}}_{\sigma^i,\sigma^c_w}(z^\tau)\Bigg] \\
&= \sigma^i(t+1|h^t)\cdot q + \sigma^i(\mathbf{d}|h^t)\cdot\nu_{t+1}(\pi(h^{t+1}),\theta,\psi(h^{t+1},\sigma^i,\sigma^c_w)) \\
&\qquad\qquad + \sigma^i(\mathbf{d}|h^t)\cdot\delta\cdot\Bigg[\sigma^i(t+2|h^{t+1})\cdot U_{t+2}(z^{t+2},\theta,\sigma^i,\sigma^c_w) \\
&\qquad\qquad + \sigma^i(\mathbf{d}|h^{t+1})\cdot\sum_{\tau=t+3}^{T} U_{t+2}(z^\tau,\theta,\sigma^i,\sigma^c_w)\cdot\mathbb{P}^{h^{t+2}}_{\sigma^i,\sigma^c_w}(z^\tau)\Bigg] \\
&= \sigma^i(t+1|h^t)\cdot q + \sigma^i(\mathbf{d}|h^t)\cdot\nu_{t+1}(\pi(h^{t+1}),\theta,\psi(h^{t+1},\sigma^i,\sigma^c_w)) \\
&\qquad\qquad + \sigma^i(\mathbf{d}|h^t)\cdot\delta\cdot\Bigg[\sigma^i(t+2|h^{t+1})\cdot\widehat{U}(t+2,h^{t+1},\theta,\sigma^i,\sigma^c_w) \\
&\qquad\qquad + \sigma^i(\mathbf{d}|h^{t+1})\cdot\widehat{U}(\mathbf{d},h^{t+1},\theta,\sigma^i,\sigma^c_w)\Bigg] \\
&= \sigma^i(t+1|h^t)\cdot q + \sigma^i(\mathbf{d}|h^t)\cdot\nu_{t+1}(\pi(h^{t+1}),\theta,\psi(h^{t+1},\sigma^i,\sigma^c_w)) \\
&\qquad\qquad + \sigma^i(\mathbf{d}|h^t)\cdot\delta\cdot\vec{U}(h^{t+1},\theta,\sigma^i,\sigma^c_w) \\
&= \sigma^i(t+1|h^t)\cdot q + \sigma^i(\mathbf{d}|h^t)\cdot\theta_q\underbrace{\sum_{\tau=t+2}^{T}\phi^{\tau-t-1}\cdot\mathbb{P}^{h^{t+1}}_{\sigma^i,\sigma^c_w}(z^\tau)\cdot q}_{\text{savoring or dread experienced at date } t+1} \\
&\qquad\qquad + \sigma^i(\mathbf{d}|h^t)\cdot\delta\cdot\vec{U}(h^{t+1},\theta,\sigma^i,\sigma^c_w) \\
&= \sum_{j=0}^{T-t-1}\delta^j\cdot\left[\theta_q\sum_{\tau=t+j+1}^{T}\phi^{\tau-t-j-1}\cdot\mathbb{P}^{h^t}_{\sigma^i,\sigma^c_w}(z^\tau) + (1-\theta_q)\cdot\mathbb{P}^{h^t}_{\sigma^i,\sigma^c_w}(z^{t+j+1})\right]\cdot q \\
&= \sum_{\tau=t+1}^{T}\left[\theta_q\sum_{j=0}^{\tau-t-1}\delta^j\phi^{\tau-t-j-1}\cdot\mathbb{P}^{h^t}_{\sigma^i,\sigma^c_w}(z^\tau)\right]\cdot q + \sum_{j=0}^{T-t-1}\left[(1-\theta_q)\delta^j\cdot\mathbb{P}^{h^t}_{\sigma^i,\sigma^c_w}(z^{t+j+1})\right]\cdot q \\
&= \sum_{\tau=t+1}^{T}\mathbb{P}^{h^t}_{\sigma^i,\sigma^c_w}(z^\tau)\cdot\left[\theta_q\sum_{j=0}^{\tau-t-1}\delta^{\tau-t-j-1}\phi^j + (1-\theta_q)\delta^{\tau-t-1}\right]\cdot q \\
&= \sum_{\tau=t+1}^{T}\mathbb{P}^{h^t}_{\sigma^i,\sigma^c_w}(z^\tau)\cdot f_q(\tau-t)\cdot q
\end{aligned}
$$

The eighth equality holds by iterating forward and replacing $\vec{U}(h^{T-1},\theta,\sigma^i,\sigma^c_w)$ with q. □

Proof of Proposition 6.1.

Fix some arbitrary $T>2$, $\theta\in\text{int}(\Theta)$, $\boldsymbol{\sigma}\in\boldsymbol{\Delta}_K$, and $q<0$. Let θ_q and f_q be defined as in Lemma 8.1. For all $\sigma^i\in\text{DP}$, let $p^1_{\sigma^i}$ $p^2_{\sigma^i}$ be defined as in the proof of Lemma 8.2. For all $t\in\{0,...,T-2\}$,

we have $p^1_{\sigma^i_1}(t) = t+1$, $p^2_{\sigma^i_1}(t) = t+2$, and $p^1_{\sigma^i_T}(t) = p^2_{\sigma^i_T}(t) = T$. Let $\underline{\theta}_{\mathcal{D}} := (1-\delta)/\phi$, and $\overline{\theta}_{\mathcal{D}} := (1-\delta^{T-1})/\sum_{j=1}^{T-1}\delta^{T-j-1}\phi^j$. If $\theta_{\mathcal{D}} < \underline{\theta}_{\mathcal{D}}$, then $f_q(1) > f_q(2)$. If $\theta_{\mathcal{D}} > \overline{\theta}_{\mathcal{D}}$, then $f_q(T) > f_q(1)$. By Lemma 8.1, these cannot hold simultaneously, so $\overline{\theta}_{\mathcal{D}} \geq \underline{\theta}_{\mathcal{D}}$.

- Suppose that $\theta_{\mathcal{D}} < \underline{\theta}_{\mathcal{D}}$. Then, $f_q(1) > f_q(2)$, so by Lemma 8.1 *(5)* f_q is strictly decreasing. By the proof of Proposition 5.1, we have $r^K_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{T\}$. By Lemma 8.3, for all $t \in \{0, ..., T-2\}$, a plan σ^i maximizes $\vec{U}(h^t, \theta, \cdot)$ iff $\mathbb{P}^{h^t}_{\sigma^i, \sigma^c_w}(z^T) = 1$. Clearly, $\mathbb{P}^{h^t}_{\sigma^i_T}(z^T) = 1$ for all $t \in \{0, ..., T-2\}$, so $\sigma^i_T \in \mathrm{UP}(\theta, \sigma^c_w)$. By Lemma 2.1, $\sigma^i_T \in \mathrm{RP}(\theta, \sigma^c_w)$. By Lemma 8.2, for all $t \in \{0, ..., T-2\}$ we have:

$$\widehat{U}(t+1, h^t, \theta, \sigma^i_T) = q = f_q(1) \cdot q < f_q(T-t) \cdot q = f_q(p^1_{\sigma^i_T}(t) - t) \cdot q = \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_T)$$

  Since the inequality is strict, σ^i_T induces no ties. Therefore, by Corollary 2.2, $|\mathrm{RP}(\theta, \sigma^c_w)| = 1$. But then, by Lemma 2.1, $\mathrm{RP}(\theta, \sigma^c_w) = \mathrm{UP}(\theta, \sigma^c_w)$.

- Suppose that $\theta_{\mathcal{D}} > \overline{\theta}_{\mathcal{D}}$. Then, $f_q(T) > f_q(1)$, so by Lemma 8.1 *(5)* $\arg\min_{n \in \mathbb{T}} f_q(n) = \{1\}$. By the proof of Proposition 5.1, we have $r^K_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{1\}$. By Lemma 8.3, for all $t \in \{0, ..., T-2\}$, a plan σ^i maximizes $\vec{U}(h^t, \theta, \cdot)$ iff $\mathbb{P}^{h^t}_{\sigma^i, \sigma^c_w}(z^{t+1}) = 1$. Clearly, $\mathbb{P}^{h^t}_{\sigma^i_1}(z^{t+1}) = 1$ for all $t \in \{0, ..., T-2\}$, so $\sigma^i_1 \in \mathrm{UP}(\theta, \sigma^c_w)$. By Lemma 2.1, $\sigma^i_1 \in \mathrm{RP}(\theta, \sigma^c_w)$. By Lemma 8.2, for all $t \in \{0, ..., T-2\}$ we have:

$$\widehat{U}(t+1, h^t, \theta, \sigma^i_1) = q = f_q(1) \cdot q > f_q(2) \cdot q = f_q(p^2_{\sigma^i_1}(t) - t) \cdot q = \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_1)$$

  Since the inequality is strict, σ^i_1 induces no ties. Therefore, by Corollary 2.2, $|\mathrm{RP}(\theta, \sigma^c_w)| = 1$. But then, by Lemma 2.1, $\mathrm{RP}(\theta, \sigma^c_w) = \mathrm{UP}(\theta, \sigma^c_w)$.

- Suppose that $\underline{\theta}_{\mathcal{D}} < \theta_{\mathcal{D}} < \overline{\theta}_{\mathcal{D}}$. Then, $f_q(T) < f_q(1) < f_q(2)$, so by Lemma 8.1 *(5)* $\arg\min_{n \in \mathbb{T}} f_q(n) = \{T\}$. By the proof of Proposition 5.1, we have $r^K_{\varnothing}(\theta, \sigma^c_w) = \{T\}$. By Lemma 8.3, a plan σ^i maximizes $\vec{U}(\varnothing, \theta, \cdot)$ iff $\mathbb{P}^{\varnothing}_{\sigma^i, \sigma^c_w}(z^T) = 1$, and maximizes $\vec{U}(h^{T-2}, \theta, \cdot)$ iff $\mathbb{P}^{h^{T-2}}_{\sigma^i, \sigma^c_w}(z^{T-1}) = 1$. These requirements are incompatible, so $\mathrm{UP}(\theta, \sigma^c_w) = \emptyset$. By Lemma 8.2, for all $t \in \{0, ..., T-2\}$ we have:

$$\widehat{U}(t+1, h^t, \theta, \sigma^i_1) = q = f_q(1) \cdot q > f_q(2) \cdot q = f_q(p^2_{\sigma^i_1}(t) - t) \cdot q = \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_1)$$

  For all $t \in \{0, ..., T-2\}$, we have $\mathrm{supp}\ \sigma^i_1(\cdot|h^t) = \{t+1\} = r_{h^t}(\theta, \sigma^c_w)$. Therefore, $\sigma^i_1 \in \mathrm{RP}(\theta, \sigma^c_w)$. Since the above inequality is strict, σ^i_1 induces no ties. Therefore, by Corollary 2.2, $|\mathrm{RP}(\theta, \sigma^c_w)| = 1$. □

Proof of Proposition 6.2.

Fix some arbitrary $T > 2$, $\theta \in \mathrm{int}(\Theta)$, $\boldsymbol{\sigma} \in \boldsymbol{\Delta}_K$, and $q > 0$. Let θ_q and f_q be defined as in Lemma 8.1. For all $\sigma^i \in \mathrm{DP}$, let $p^1_{\sigma^i}$ and $p^2_{\sigma^i}$ be defined as in the proof of Lemma 8.2. For all $t \in \{0, ..., T-2\}$, we have $p^1_{\sigma^i_1}(t) = t+1$, $p^2_{\sigma^i_1}(t) = t+2$, and $p^1_{\sigma^i_T}(t) = p^2_{\sigma^i_T}(t) = T$. Let $\underline{\theta}_{\mathcal{S}} := (1-\delta)/\phi$, and define:

$$\overline{\theta} := \frac{\delta^{T-2} - \delta^{T-1}}{\delta^{T-2} - \delta^{T-1} + \sum_{j=0}^{T-1}\delta^{T-j-1}\phi^j - \sum_{j=0}^{T-2}\delta^{T-j-2}\phi^j} \qquad \overline{\theta}_{\mathcal{S}} := \begin{cases} \overline{\theta} & \text{if} \quad \overline{\theta} > 0 \\ \infty & \text{if} \quad \overline{\theta} \leq 0 \end{cases}$$

If $\theta_{\mathcal{S}} < \underline{\theta}_{\mathcal{S}}$, then $f_q(1) > f_q(2)$. If $\theta_{\mathcal{S}} > \overline{\theta}_{\mathcal{S}}$, then $f_q(T) > f_q(T-1)$. By Lemma 8.1, these cannot hold simultaneously, so $\overline{\theta}_{\mathcal{S}} \geq \underline{\theta}_{\mathcal{S}}$.

- Suppose that $\theta_{\mathcal{S}} < \underline{\theta}_{\mathcal{S}}$. Then, $f_q(1) > f_q(2)$, so by Lemma 8.1 *(5)* f_q is strictly decreasing. By the proof of Proposition 5.2, we have $r^K_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{1\}$. By Lemma 8.3, for all $t \in \{0, ..., T-2\}$, a plan σ^i maximizes $\vec{U}(h^t, \theta, \cdot)$ iff $\mathbb{P}^{h^t}_{\sigma^i, \sigma^c_w}(z^{t+1}) = 1$. Clearly, $\mathbb{P}^{h^t}_{\sigma^i_1}(z^{t+1}) = 1$ for all $t \in \{0, ..., T-2\}$, so $\sigma^i_1 \in \text{UP}(\theta, \sigma^c_w)$. By Lemma 2.1, $\sigma^i_1 \in \text{RP}(\theta, \sigma^c_w)$. By Lemma 8.2, for all $t \in \{0, ..., T-2\}$ we have:

$$\widehat{U}(t+1, h^t, \theta, \sigma^i_1) = q = f_q(1) \cdot q > f_q(2) \cdot q = f_q(p^2_{\sigma^i_1}(t) - t) \cdot q = \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_1)$$

  Since the inequality is strict, σ^i_1 induces no ties. Therefore, by Corollary 2.2, $|\text{RP}(\theta, \sigma^c_w)| = 1$. But then, by Lemma 2.1, $\text{RP}(\theta, \sigma^c_w) = \text{UP}(\theta, \sigma^c_w)$.

- Suppose that $\theta_{\mathcal{S}} > \overline{\theta}_{\mathcal{S}}$. Then, $f_q(T) > f_q(T-1)$, so by Lemma 8.1 *(5)* f_q is strictly increasing over $\mathbb{T}$. By the proof of Proposition 5.2, we have $r^K_{\varnothing}(\theta, \boldsymbol{\sigma}) = \{T\}$. By Lemma 8.3, for all $t \in \{0, ..., T-2\}$, a plan σ^i maximizes $\vec{U}(h^t, \theta, \cdot)$ iff $\mathbb{P}^{h^t}_{\sigma^i, \sigma^c_w}(z^T) = 1$. Clearly, $\mathbb{P}^{h^t}_{\sigma^i_T}(z^T) = 1$ for all $t \in \{0, ..., T-2\}$, so $\sigma^i_T \in \text{UP}(\theta, \sigma^c_w)$. By Lemma 2.1, $\sigma^i_T \in \text{RP}(\theta, \sigma^c_w)$. By Lemma 8.2, for all $t \in \{0, ..., T-2\}$ we have:

$$\widehat{U}(t+1, h^t, \theta, \sigma^i_T) = q = f_q(1) \cdot q < f_q(T-t) \cdot q = f_q(p^1_{\sigma^i_T}(t) - t) \cdot q = \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_T)$$

  Since the inequality is strict, σ^i_T induces no ties. Therefore, by Corollary 2.2, $|\text{RP}(\theta, \sigma^c_w)| = 1$. But then, by Lemma 2.1, $\text{RP}(\theta, \sigma^c_w) = \text{UP}(\theta, \sigma^c_w)$.

- Suppose that $\underline{\theta}_{\mathcal{S}} < \theta_{\mathcal{S}} < \overline{\theta}_{\mathcal{S}}$. Then, $f_q(T) < f_q(T-1)$ and $f_q(1) < f_q(2)$, so $1, T \notin \arg\max_{n \in \mathbb{T}} f_q(n)$. By the proof of Proposition 5.2, $1, T \notin r^K_{\varnothing}(\theta, \sigma^c_w)$. Suppose that σ^i_* maximizes $\vec{U}(\varnothing, \theta, \cdot)$. By Lemma 8.3, $\mathbb{P}^{\varnothing}_{\sigma^i_*, \sigma^c_w}(z^T) = 0$, so there exists some $\tau \in \{1, ..., T-1\}$ such that $\mathbb{P}^{h^{\tau-1}}_{\sigma^i_*, \sigma^c_w}(z^\tau) \geq \mathbb{P}^{\varnothing}_{\sigma^i_*, \sigma^c_w}(z^\tau) > 0$. But then, σ^i_* does not maximize $\vec{U}(h^{\tau-1}, \theta, \cdot)$, so $\sigma^i_* \notin \text{UP}(\theta, \sigma^c_w)$. Therefore, $\text{UP}(\theta, \sigma^c_w) = \emptyset$. Let $n^* := \min\{n \in \mathbb{N} \colon f_q(n+1) < 1\}$. Since $f_q(1) < f_q(2)$, we have $n^* > 1$. For all $t \in \{0, ..., T-2\}$, we have $p^1_{\sigma^i_{\text{mod}(n^*)}}(t) \leq t + n^*$. For all $t \in \{0, ..., T-2\}$ such that $\sigma^i_{\text{mod}(n^*)}(t+1|h^t) = 1$, we have $p^2_{\sigma^i_{\text{mod}(n^*)}}(t) = t + n^* + 1$. Let $\theta^*_{\mathcal{S}} := (1 - \delta^{n^*-1}) / \sum_{j=1}^{n^*-1} \delta^{n^*-j-1} \phi^j$. Suppose that $\theta_{\mathcal{S}} \neq \theta^*_{\mathcal{S}}$, so $f_q(n^*) > f_q(1)$. By Lemma 8.2, for all $t \in \{0, ..., T-2\}$ such that $\sigma^i_{\text{mod}(n^*)}(\mathbf{d}|h^t) = 1$, we have:

$$\widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_{\text{mod}(n^*)}) = f_q(p^1_{\sigma^i_{\text{mod}(n^*)}}(t) - t) \cdot q > f_q(1) \cdot q = q = \widehat{U}(t+1, h^t, \theta, \sigma^i_{\text{mod}(n^*)})$$

  For all $t \in \{0, ..., T-2\}$ such that $\sigma^i_{\text{mod}(n^*)}(t+1|h^t) = 1$, we have:

$$\widehat{U}(t+1, h^t, \theta, \sigma^i_{\text{mod}(n^*)}) = q = f_q(1) \cdot q > f_q(n^*+1) \cdot q = f_q(p^2_{\sigma^i_{\text{mod}(n^*)}}(t) - t) \cdot q = \widehat{U}(\mathbf{d}, h^t, \theta, \sigma^i_{\text{mod}(n^*)})$$

  For all $t \in \{0, ..., T-2\}$, we have supp $\sigma^i_{\text{mod}(n^*)}(\cdot|h^t) = r_{h^t}(\theta, \sigma^c_w)$. Therefore, $\sigma^i_{\text{mod}(n^*)} \in \text{RP}(\theta, \sigma^c_w)$. Since the above inequality is strict, $\sigma^i_{\text{mod}(n^*)}$ induces no ties. Therefore, by Corollary 2.2, $|\text{RP}(\theta, \sigma^c_w)| = 1$. □